

The emergence and cultural persistence of ineffective solutions

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Abstract

Through cumulative cultural evolution, human populations discover novel solutions that allow them to solve complex problems. However, populations also reliably develop and sustain practices that are costly yet appear to have no effect on their stated outcome, such as bloodletting, divination, and malicious sorcery. Here, we show that ineffective, costly solutions can develop and be maintained because of their subjective appeal. In a pre-registered transmission-chain experiment, participants were tasked with developing a solution to a problem over several trials, after which they passed instructions onto the subsequent generation. The visual appearance of the experimental task was manipulated to vary the extent to which cognitive biases could interfere with participants' expectations of efficacy, enabling a direct test of the effect of cognitive appeal on cumulative cultural evolution. Participants in the low-appeal treatment (n=240) progressively developed more efficient solutions across generations. Participants in the high-appeal treatment (n=240), in contrast, produced costlier solutions with no increase in efficiency over the course of the experiment. These results suggest that, even under conditions favorable to cumulative cultural evolution, a form of cognitive appeal can alone prevent the process and promote the emergence and cultural persistence of ineffective solutions.

Introduction

Cumulative cultural evolution appears critical for humans' success¹⁻³. Through the accumulation of cultural innovations over many generations, populations have developed solutions that no single individual could have invented, allowing us to inhabit new environments and achieve feats unparalleled in the animal world, from space exploration to industrial-scale agriculture. Several experiments have shown that cumulative cultural evolution occurs through instrumental, payoff-biased social learning, which allows the gradual accumulation of beneficial modifications over time⁴⁻⁶. Participants tasked with solving a problem, such as building a basket to carry rice or designing a wheel to roll down a track, preferentially adopt and retain solutions that yield greater expected payoffs⁷⁻⁹. When successive generations of learners select both successful demonstrators and effective-seeming solutions, cumulative cultural evolution can drive the emergence of complex, highly efficient solutions^{6,8}.

Theoretical and empirical work has shown that cumulative cultural evolution operates under a wide range of conditions, albeit with differing levels of effectiveness. When payoffs are immediate and accurate, payoff-biased social learning allows individuals to filter out less efficient solutions, gradually bringing populations closer to the optimal solution over time^{10,11}. However, when payoffs are noisy or delayed, gradual improvements are slower, as individuals struggle to accurately assess the effectiveness of solutions¹². Additionally, experimental work has demonstrated that populations tend to converge on less efficient solutions in environments that are misaligned with their prior assumptions¹³. These mitigating factors notwithstanding, cumulative cultural evolution remains powerful enough to foster the emergence of increasingly efficient solutions across diverse conditions, even when those solutions are counterintuitive and beyond individual understanding^{8,9}.

However, while cumulative cultural evolution reliably promotes the emergence of increasingly efficient solutions in experimental settings, it also appears to produce and preserve ineffective practices in the real world. Humans around the world engage in practices that seem to have no effect on their stated outcome, such as divination^{14,15}, rain magic¹⁶, malicious sorcery¹⁷, and bloodletting¹⁸. These practices are widespread and elaborate, requiring non-trivial investments of time and resources. Research at the intersection of cognitive science and anthropology suggests that these practices may develop because of systematic biases in how individuals evaluate solutions. These biases are examples of "content biases" that can influence cultural evolution¹⁹. For example, prior research showed that participants in the United States

and Brazil evaluated rituals as more effective when they involved features like repetition or specific times, despite the rituals having no effect on their intended outcome²⁰.

Of the features that can bias evaluations of potential solutions, one of the most widespread seems to be a visual correspondence between an intervention and its target. Anthropologists have long observed that people often use magic and folk medicines that visually resemble the desired outcome²¹. Well-known examples include voodoo dolls or effigies, in which a practitioner tries affecting another person by manipulating an object that resembles them, as well as the doctrine of signatures, which holds that herbs resembling specific body parts can be used to treat ailments affecting those parts^{22,23}. Such correspondences have been hypothesized to tap into widespread cognitive biases^{24,25}.

Here, we hypothesize that subjective appeal—in this case, the potential for similarity between an intervention and its target—can drive the emergence and persistence of ineffective solutions even in conditions otherwise conducive to cumulative cultural evolution. To test this hypothesis, we conducted a pre-registered experiment in which participants were tasked with maximizing their score by allocating nutrients to nine plants in a virtual field (Figure 1A). Nutrients came in three colors of different costs—blue (2 units), yellow (6 units), and red (10 units)—and participants could allocate up to sixteen nutrients across the nine plants. This resulted in a vast solution space, with billions of distinct configurations. The field’s output varied as a function of how nutrients were allocated across plants, although scores were noisy through the inclusion of a random component (see Methods section). The optimal solution was to provide each plant with a single nutrient of the lowest cost (the blue nutrient).

Participants were organized into experimental chains of four generations, with each generation consisting of three participants (Figure 1B). Each participant completed six trials and received feedback on their score after each one. After their final trial, participants wrote a piece of advice for the next generation. From the second generation onward, participants were shown the final scores and written advice of three individuals from the previous generation and were asked to choose one as their cultural demonstrator before beginning their first trial.

We implemented two experimental treatments that differed in the degree of visual similarity between the features of the plants and those of the nutrients. In the low-appeal treatment (n=240), the plants had different shapes but were all gray in color (Figure 1A). In the high-appeal treatment (n=240), in contrast, plants were additionally assigned one of six colors: plant colors were either primary colors, matching the colors of the nutrients (blue, yellow, red),

or secondary colors that can typically be created by mixing the primary colors (green, purple, orange). Thus, in this treatment, the nutrients fed to the plants could resemble the plants themselves, potentially allowing for more visually appealing solutions than in the low-appearance treatment. Importantly, the size of the solution space was kept equivalent across experimental treatments by pairing each plant shape with a unique color (Figure 1A).

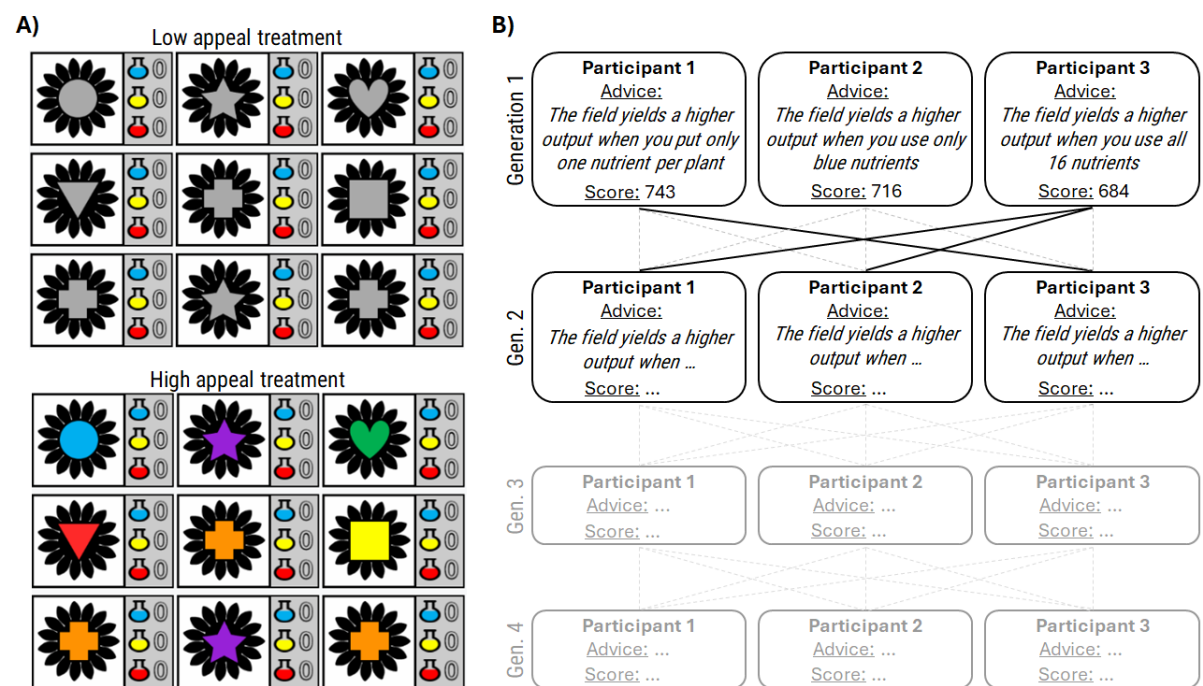


Figure 1. Problem-Solving Task and Experimental Design. A) Plant-Feeding Task: Participants were presented with six uniquely shaped flowers within a field of nine plants. The plants appeared either uniformly gray (low-appearance treatment) or in various colors, with each shape paired to a specific color (high-appearance treatment). Participants had access to three different nutrient types and could use up to 16 nutrients across the entire field. After each trial, participants received feedback on the score their solution generated. B) Participants were organized into experimental chains consisting of four generations, with three participants per generation. After engaging with the task for six trials, each participant wrote a piece of advice for participants in the subsequent generation. From Generation 2 onward, participants could select one cultural demonstrator based on that individual's written advice and score in their final trial before solving the task and producing their own piece of advice for participants in the following generation.

Under these conditions, rates of cumulative cultural evolution should be comparable across treatments if cognitive biases do not influence participants' expectations of efficacy. Conversely, if changes in the visual appearance of stimuli interfere with these expectations, we would expect rates of cumulative cultural evolution to differ between treatments. To test this, we compared the score and cost of culturally transmitted solutions—that is, the solutions produced on the sixth trial—across generations in the low-appearance and high-appearance treatments. We predicted that the score of transmitted solutions would increase over generations in the

low-appeal treatment and that this increase would exceed any change observed in the high-appeal treatment ([link to the preregistration](#)). Additionally, we predicted that participants in the low-appeal treatment would rely on payoff information to a greater extent when selecting their cultural demonstrator than participants in the high-appeal treatment.

Results

In the low-appeal treatment, the score of the transmitted solutions increased over the four generations (Figure 2A), while the cost of these solutions progressively decreased (Figure 2B, see Methods for details on score and cost calculation). In the high-appeal treatment, in contrast, neither the score of the transmitted solutions (Figure 2A) nor the cost of these solutions (Figure 2B) changed significantly over the course of the experiment. In the low-appeal treatment, scores increased by an estimated 17.23 points with each additional generation ($SE = 4.95$, $p < 0.001$), whereas in the high-appeal treatment, the change in score across generations was statistically indistinguishable from zero ($\beta=2.72$, $SE = 4.85$; Supplementary Table A1).

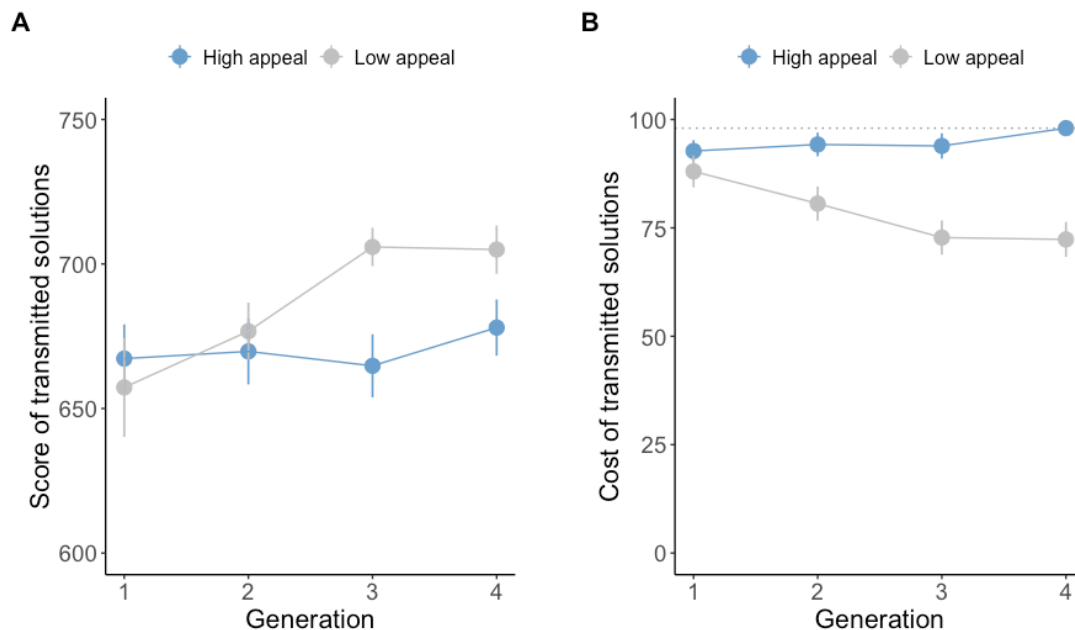


Figure 2. Score and cost of culturally transmitted solutions across generations. A) In the low-appeal treatment (gray line), participants showed gradual improvements in solution scores across generations, whereas scores in the high-appeal treatment (blue line) did not change significantly. B) Participants in the low-appeal treatment progressively reduced the cost of their solutions, while those in the high-appeal treatment showed no statistically significant change over time. The dotted line represents the cost of the mix-and-match solution (see Figure 5 for an illustration).

Among the three variables that affected solution scores (number of plants treated, number of nutrients used, and the type of nutrients used; see Methods), only the type of

nutrients differed between treatments, with the number of blue nutrients increasing across generations in the low-appeal treatment (Generation: $\beta = -0.11$, $SE = 0.17$, $p = ns$; Low appeal: $\beta = -0.54$, $SE = 0.87$, $p = ns$; Generation $\times$ Low appeal: $\beta = 0.95$, $SE = 0.25$, $p < 0.001$; see Supplementary Figure A1 and Table A2 for details).

Contrary to our second prediction, we observed no differences between treatments in social learning strategies. Participants in both treatments relied on payoff information when choosing among the three available cultural models, selecting the highest-scoring demonstrators 72% of the time in the low-appeal treatment and 78% of the time in the high-appeal treatment. The mean agreement among learners, calculated as the pairwise proportion of learners within a chain who selected the same demonstrator, was also comparable between treatments (67.2% and 66.1%, respectively; see Supplementary Figure A2), as was the probability of a participant implementing the inherited solution in their first trial (14.4% and 18.3%, respectively; see Supplementary Figure A3.A and Table A3). However, the similarity between the solution participants inherited and the solution they used in their first trial was higher in the high-appeal treatment than in the low-appeal treatment overall (Generation: $\beta = 0.47$, $SE = 0.66$, $p = 0.48$; Low appeal: $\beta = 6.61$, $SE = 3.00$, $p = 0.028$; Generation $\times$ Low appeal: $\beta = -1.45$, $SE = 0.93$, $p = 0.118$; see Supplementary Figure A3.b and Table A3 for details).

To understand why cumulative performance differed between treatments despite similar social learning strategies, we conducted exploratory analyses comparing exploration and cultural transmission patterns. Results from these analyses show that participants explored the solution space more extensively in the low-appeal treatment than in the high-appeal treatment. On average, they produced 5.43 unique solutions out of 6 trials in the low-appeal treatment, compared to 4.61 in the high-appeal treatment. Across all experimental replicates, 77.8% of solutions were unique in the low-appeal treatment, compared to only 52.4% in the high-appeal treatment. Figure 3 shows the cumulative number of unique solutions across trials, illustrating that exploration was broader in the low-appeal condition. Reduced exploration was also apparent in analyses of within-participant solution diversity across generations, measured by the pairwise distance between solutions in six trials. Within-participant diversity tended to decrease across generations in both treatments, but remained overall lower in the high-appeal treatment (Generation: $\beta = -1.59$, $SE = 0.34$, $p < 0.001$; Low appeal: $\beta = 4.35$, $SE = 1.44$, $p = 0.003$; Generation $\times$ Low appeal: $\beta = 0.05$, $SE = 0.48$, $p = 0.917$; see Supplementary Figure A4 and Table A4 for details). Additionally, within-chain and between-chain diversity was lower

in the high-appeal treatment than in the low-appeal treatment (see Supplementary Figure A5 and Table A5 for details).

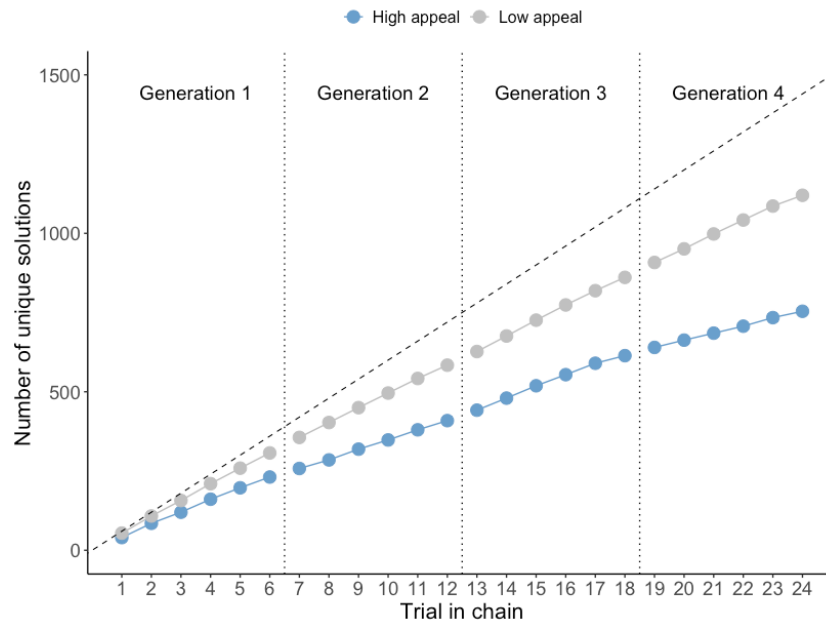


Figure 3. Exploration across trials. Cumulative number of unique solutions (i.e., nutrient-to-plant allocation patterns) across trials in each treatment. The dashed black line indicates the theoretical upper bound of exploration, assuming every participant submitted a solution not previously produced in this treatment. The gray line shows the cumulative number of unique solutions observed in the low-appeal treatment, while the blue line shows the high-appeal treatment.

Analyses of the solutions produced in the sixth trial—that is, the solutions transmitted between generations—reveal important differences between treatments. In the high-appeal treatment, the most common solution accounted for about 16% of all transmitted solutions (38 of 240, Figure 4), and the two most common solutions together accounted for about 22% of transmitted solutions (53 of 240). In the low-appeal treatment, no single solution was observed more than seven times, and the two most common solutions accounted for about 5% of all transmitted solutions (13 of 240). Overall, solutions observed more than once represented about 42.5% of all transmitted solutions in the high-appeal treatment (102 of 240 solutions across 14 unique solutions), compared with about 15% in the low-appeal treatment (36 of 240 solutions across 13 unique solutions).

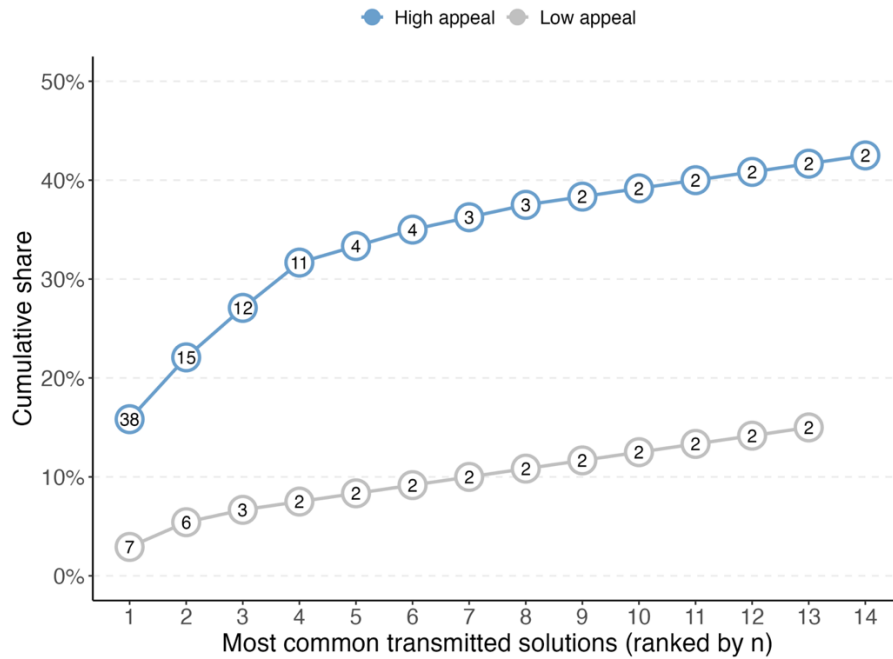


Figure 4. Cumulative share of most common transmitted solutions. Cumulative proportion of transmitted solutions ranked by frequency in each treatment. In the high-appeal condition, a relatively small number of solutions account for a substantial share of transmitted solutions. In the low-appeal condition, transmitted solutions are more evenly distributed across a broader range of configurations, with no single solution dominating to the same extent. Numbers inside the circles indicate the number of times each solution was transmitted. Only solutions observed more than once are represented.

Figure 5 illustrates how the most common solutions visually differed between treatments. In the low-appeal treatment, the two most common solutions involved only blue nutrients: the most common used 16 blue nutrients, while the second most common—which was the optimal solution—used 9 blue nutrients, one per plant (Figure 5). In the high-appeal treatment, the most common solutions followed a costly 'mix-and-match' strategy: participants matched nutrient colors to flower colors whenever possible, and when exact matches were unavailable, they combined nutrient colors based on primary-color mixing (e.g., mixing blue and red nutrients for a purple flower; Figure 5). These mix-and-match solutions cost approximately five times more than the optimal solution.

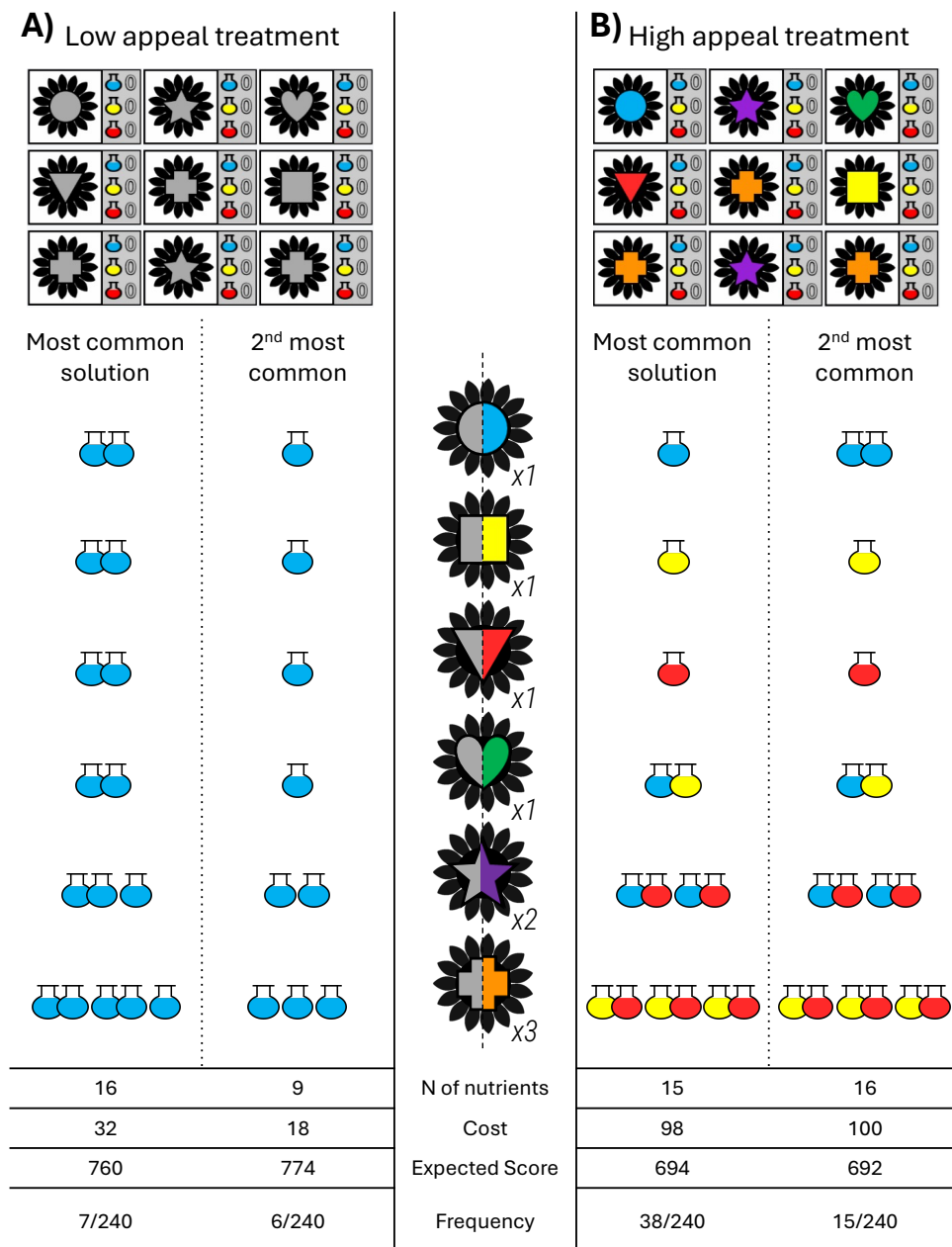


Figure 5. Most common culturally transmitted solutions. The central column shows the composition of the field, alongside the nutrient configurations used in the most common solutions for each treatment. Note that the star/violet and orange/cross plants appeared twice and three times, respectively, in the field. A) The two most common culturally transmitted solutions in the low-appeal treatment exclusively involve blue nutrients. The most common solution in this treatment uses 16 blue nutrients, the most efficient option when all nutrients are used. The second most common solution involves 9 blue nutrients—one for each plant—which is the optimal solution. B) In the high-appeal treatment, the two most common solutions use a mix-and-match strategy, where primary-colored plants are fed with the corresponding colored nutrient, while secondary-colored plants receive a mix of colored nutrients.

Analyses of the written advice show that advice messages in the high-appeal treatment had higher semantic similarity, despite being longer, than those in the low-appeal treatment (see Supplementary Material and Supplementary Figure A7 for details). A classification of advice content indicates that participants in the high-appeal treatment transmitted advice

consistent with the transmitted feeding patterns, most often recommending the “mix-and-match” strategy throughout the experiment. By contrast, advice in the low-appeal treatment increasingly recommended favoring blue nutrients over generations, shifting from relatively uncommon in the first generation to being frequently transmitted by the end of the experiment (see Supplementary Figure A8).

Discussion

Our findings demonstrate that changes to the visual appearance of stimuli, by increasing the opportunity for cognitive biases to interfere with expectations of efficacy, can hinder cumulative cultural evolution. In the low-appeal treatment, where all plants were gray, solutions became progressively more efficient across generations. In contrast, in the high-appeal treatment, where plants were colorful, participants produced costly, ineffective solutions and showed no gradual improvement across generations.

This failure to shift toward more efficient solutions in the high-appeal treatment appears to be driven by reduced exploration both within and across participants. Our findings suggest that this lower level of exploration reflects the strong appeal of the “mix-and-match” strategy in the high-appeal treatment, wherein participants aligned nutrient colors with flower colors whenever possible or combined nutrient colors based on primary-color mixing. The appeal of this strategy is evident in several ways. First, participants in the high-appeal treatment produced fewer unique solutions, indicating a shift toward greater exploitation in the exploration–exploitation dilemma faced by participants. This pattern is consistent with the idea that cognitive biases shaped participants’ expectations of efficacy in our experiment. Second, when participants did explore, exploration was more local and more convergent in the high-appeal treatment: successive solutions produced by the same participant were more similar (smaller within-individual solution distances across the six trials), and solutions overlapped more across participants (smaller within-chain and between-chain solution distances).

Limited exploration may also have been self-reinforcing through the cultural transmission of advice. Experiments have shown that receiving a theory can channel exploration toward a specific region of the search space ^{8,26}. In our experiment, the limited variation in the theories transmitted in the high-appeal treatment, which tended to focus on the mix-and-match solution, would therefore be expected to channel exploration toward a similar region across participants, further constraining the total exploration across individuals.

Regardless of the precise mechanism, the observed reduction in variation constrains cultural evolvability—the capacity of a cultural system to generate, retain, and build upon culturally transmitted variation—by limiting the raw material on which selection operates ²⁷.

Importantly, the absence of gradual improvement across generations in the high-appeal treatment occurred even though participants relied on payoff information for selecting their cultural demonstrator, most often choosing the highest-scoring option. Payoff-biased social learning is known to be critical for the emergence of increasingly efficient solutions, which is consistent with findings from our previous version of the same experiment in which payoff information was unavailable for demonstrator choice and no cumulative improvement was observed in either treatment (see Supplementary Material). Disrupting participants' ability to identify the highest-performing cultural demonstrator would therefore be expected to hinder cumulative cultural evolution even more than what we observe here.

A likely reason that participants continued to rely on payoff information, despite the opportunity to also judge demonstrators by the appeal of their advice, is that payoff comparison was cognitively straightforward (participants compared three numerical scores), whereas comparing advice required reading and integrating several text messages. Outside the laboratory, payoff information is often noisier, delayed, or difficult to quantify, which could increase reliance on appeal. A second factor is that strong convergence in the high-appeal condition limited variation in the appeal of competing messages. Consistent with this, advice messages in the high-appeal condition were more homogeneous than in the low-appeal condition and thus less diagnostic for differentiating demonstrators, making payoffs a more informative cue.

The core feature of mix-and-match—a perceptual alignment between the intervention and the target object—mirrors patterns found in magic and folk remedies, exemplifying what anthropologists term the “law of similarity” ²¹. Such cultural patterns suggest that humans reliably develop tendencies to infer causal relationships from visual similarity. The reasons for such causal biases remain debated; proposals include that people erroneously learn causal relationships from environmental regularities²⁴ and that the principle of like-affects-like occurs frequently enough in the natural world that it proved a useful scaffold for understanding causality for our ancestors²². Whatever their developmental or evolutionary origins, such intuitions can lead individuals to overestimate causal links between features and effects, such as assuming that a plant's shape is linked to its medicinal properties or, as in our experiment, that nutrients are more effective when their colors match those of the plants to which they are

301 applied. The exposure to cultural products that harness these intuitions, such as video games,
302 may further amplify such causal associations.

303 Given the apparent functionality of many cultural practices found across much of
304 culture, researchers have often explained those that seem instrumentally ineffective, such as
305 shamanism, magic, and divination, by proposing alternative functional benefits, such as
306 enhanced confidence or social cohesion ^{28,29,30}. However, consistent with recent arguments
307 ^{14,31,32}, our results experimentally illustrate that ineffective cultural practices can be maintained
308 by subjective appeal. Our experiment investigated a single form of subjective appeal—visual
309 similarity—and the resulting cultural product was much simpler than practices like divination
310 or shamanism. We suspect that these more complex practices tap into other dimensions of
311 subjective appeal beyond those investigated here. Nevertheless, our study serves as proof of
312 concept, demonstrating that functional benefits are not required for the emergence and
313 persistence of costly ineffective cultural practices.

314 Our study has important implications for studying cumulative cultural evolution under
315 controlled conditions. Although researchers have investigated the conditions that constrain
316 cumulative cultural evolution ^{5,13,33-38}, most experiments are designed in ways that may
317 artificially inflate the potential for gradual cultural improvement. For example, with few
318 exceptions ¹², researchers typically provide participants with immediate and accurate payoffs,
319 despite many interventions in the real-world leading to payoffs that are noisier and time-
320 delayed. More importantly, the solution space in laboratory experiments is often artificially
321 constrained, limiting the opportunity for inefficient but cognitively appealing solutions to
322 compete with more efficient ones. These design choices potentially lead to an
323 underappreciation of the inefficiencies of cumulative cultural evolution as manifested both in
324 practices that are cheap and effective yet rare, such as handwashing ³⁹, and in practices that are
325 common and costly yet futile.

326 In our experiment, the high-appeal treatment created opportunities for visual similarity,
327 which we expected would bias how participants evaluated potential solutions. By embedding
328 this cognitive bias into the task environment, we created a setting in which visually appealing—
329 but costly and inefficient—solutions could plausibly emerge. This does not mean, however,
330 that the experiment was artificially tailored to produce such outcomes. First, the patterns
331 observed in the low-appeal treatment confirm that the conditions of the experiment—such as
332 task complexity, payoff noise, and number of demonstrators—were conducive to cumulative
333 cultural evolution. Second, even in the high-appeal treatment, the opportunity for cognitively

appealing solutions to emerge was limited. In real-world contexts, learners routinely encounter similarity-based cues alongside many other sources of biased inference. Outside the lab, individuals interact with a broader set of variables—such as timing and other contextual factors—that may trigger additional biases not captured in our design²⁰. This broader and more complex inferential landscape may further amplify the emergence and persistence of ineffective solutions, underscoring the importance of understanding how subjective appeal shapes the dynamics of cultural evolution across contexts and domains.

Methods

Ethics Statement. All methods were approved by the Toulouse School of Economics/Institute for Advanced Studies in Toulouse Review Board for Ethical Standards in Research (2022-11-001). All participants provided informed consent before taking part in the experiment. All individual information has been anonymized.

Participants. A total of 480 participants were recruited for the study. Participants were recruited through Prolific and filtered based on several criteria: they were required to be aged between 18 and 35 years, from the United Kingdom, United States, Ireland, Australia, Canada, or New Zealand, and to have English as their first language. Additionally, participants needed to have no issues seeing colors and maintain an approval rate of 98-100% on Prolific. Participants received a £1.2 participation fee, with an average bonus of £1.8 based on their performance.

Experimental task. The experiment relied on a plant-feeding task in which participants were presented with a virtual field containing nine plants (Figure 1A). Participants aimed to maximize the output of their virtual field across 6 trials by feeding the plants with three types of nutrients that varied in their cost: blue (2 units), yellow (6 units), and red (10 units). Participants' decisions in each trial of nutrient allocation would impact the output of their field and the cost of their solution, which affected their score for that trial. The field's output was determined by the number of plants that receive at least one nutrient and a random component. Additional nutrients on the same plant, the shapes and colors of plants, or the color of the nutrients used had no effect on the output. Put precisely, the field's output was the number of plants fed, multiplied by a uniformly distributed number between 60 and 100. Subtracting the total nutrient cost and adding a fixed score of 72 resulted in the participant's score on that trial:

$$\text{Score} = 72 + \text{Uniform}(60, 100) \times \text{Number of plants fed} - \text{Total cost of nutrients}$$

The fixed score of 72 points was added to prevent participants from having a negative score in the worst-case scenario.

Participants were limited to using a maximum of 16 nutrients across the entire field, with each plant receiving no more than 9 nutrients of a given color. Under these constraints, trial scores could range from 0 to 954. The optimal strategy (feeding each plant a single blue nutrient) yielded an average score of 774. However, in the best-case scenario, the field could produce an output of 900 (9 plants $\times$ 100), at a cost of 18, resulting in a total score of $72 + 900 - 18 = 954$. In contrast, the worst strategy (feeding a single plant 9 red and 7 yellow nutrients) yielded an average score of 20. In the worst-case scenario, this approach could produce a field output of just 60 at a cost of 132, resulting in a total score of $72 + 60 - 132 = 0$.

Participants were informed about the random component affecting their score. At the end of each trial, they were shown their score for that trial. However, they were not told the exact function governing the field's output and the costs of the nutrients.

Treatments. The study implemented two treatments: low-appeal and high-appeal (see Figure 1A). A total of 480 participants were equally and randomly assigned to one of the two treatment conditions: high-appeal ($n=240$) and low-appeal ($n=240$). Each condition consisted of 20 transmission chains, each comprising four generations of three participants, for a total of 12 participants per chain. In the low-appeal treatment, all plants were gray and exhibited one of six distinct shapes: circle, star, heart, triangle, cross, and square (see Figure 1A, upper figure). The high-appeal treatment featured the same shapes but in distinct colors: the round shape was blue, the star was purple, the heart was green, the triangle was red, the cross was orange, and the square was yellow (see Figure 1A, lower figure). All participants in each treatment saw the same fixed arrangement of the plants. In both treatments, the shapes and colors of the plants were purely aesthetic differences and did not influence the output of the field.

Cultural transmission. The participants engaged with the task over six trials. Each participant's last solution and its corresponding score for that trial were passed onto the next participant in their chain. In addition to transmitting their solutions, each participant was required to provide an accompanying piece of advice for the next participant in their chain (see Figure 1B). The advice was pre-formatted to begin with the phrase "*The field yields a higher output when...*" to ensure comparability across responses. Before starting the task, participants were provided with three pieces of advice from participants in the previous generation, along with their score from the last trial, and could select one participant as cultural demonstrator.

Upon selecting a cultural demonstrator, participants were provided with the chosen piece of advice along with the participants' solution (i.e., which nutrients were provided to each plant) for the duration of the experiment.

Statistical analysis.

Score and cost of transmitted solutions. We used the transmitted solutions of each of the 480 participants as the unit of analysis in our models. The response variable was each participant's expected score—that is, the score excluding the effect of the random component—to reduce the impact of artificial noise on the statistical analyses. To compare how the score changed over the course of the experiment across the two treatment conditions, we first conducted two separate models using subsets of the data for each treatment ($n=240$ for each, $n_{\text{chain}}=20$). To assess how the expected score changed across generations within a given treatment condition, we used a multilevel regression model with an interaction term between generation and treatment as a fixed effect, with the pooled data (high-appeal: $n=240$, low-appeal: $n=240$, $n_{\text{chain}}=40$). We repeated those three models using participants' costs as the response variable. In all models, we accounted for chain-specific random effects.

Social learning strategies. The response variable was a binary indicator of whether participants selected the highest-scoring demonstrator among the three available cultural models. We used a multilevel model with treatment as a fixed effect and included chain and generation as random effects.

Distance measure. For analyses comparing configurations, we used a distance measure based on representing each solution as a vector of nutrient counts, i.e. an ordered sequence of 27 non-negative integers recording how many blue, yellow, and red nutrients each of the nine plants received. This representation is consistent across all solutions, allowing direct numerical comparison. To quantify the difference between solutions, we computed the Manhattan distance between their respective vectors. We computed the mean pairwise distance as the average distance between all possible pairs of solutions in the set. Higher values of mean pairwise distance indicate that solutions in the set are more dispersed and varied, while lower values indicate convergence toward similar solutions. We used this measure to quantify within-individual exploration across trials, as well as convergence of solutions within and between transmission chains.

All data processing was performed using the R software environment⁴⁰, and we conducted multilevel regression analyses with the lmer function from the lme4 package in R⁴¹.

Pre-registration. The experiment was pre-registered on AsPredicted prior to data collection ([link](#)).

Data and Code Availability. The data and scripts for the analyses are publicly available at: <https://github.com/aseyq/inefficient-solutions>

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Supplementary material

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Preregistered Results and Additional Analyses

A) Scores			
	Low-Appeal only	High-Appeal only	Pooled
(Intercept)	643.17 *** (13.94)	663.15 *** (13.60)	663.15 *** (13.77)
Generation	17.23 *** (4.95)	2.72 (4.85)	2.72 (4.90)
Low Appeal			-19.98 (19.48)
Generation x Low Appeal			14.51 * (6.93)
B) Costs			
(Intercept)	92.17 *** (5.40)	90.85 *** (3.09)	90.85 *** (4.40)
Generation	-5.49 *** (1.50)	1.55 (1.12)	1.55 (1.32)
Low Appeal			1.32 (6.22)
Generation x Low Appeal			-7.04 *** (1.87)
Num. obs.	240	240	480
Num. groups: chain	20	20	40

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Table A1. Multilevel model estimates for *expected scores (A)* and *costs (B)* of culturally transmitted solutions across treatments. Separate models are reported for the low-appeal condition, high-appeal condition, and pooled data to compare changes in expected scores/costs over generations, with random intercepts controlling for transmission *chain* structure of the experiment. Standard errors are shown in parentheses and statistical significance is denoted by asterisks (*: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$).

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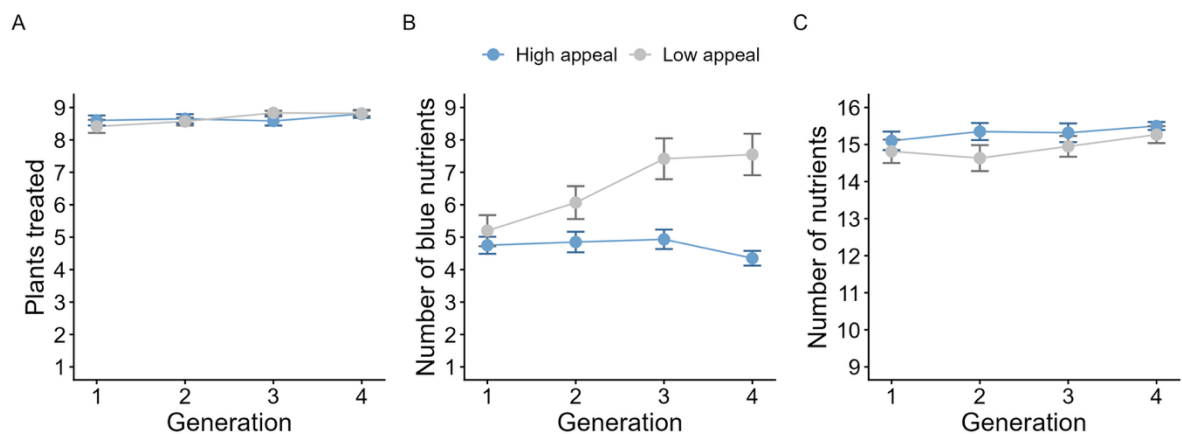


Figure A1. Patterns of nutrient use. (A) Mean number of plants treated (i.e., plants receiving at least one nutrient). (B) Mean number of blue nutrients used. (C) Mean total number of nutrients used. Error bars represent ± 1 SE. Across treatments, participants consistently fed most plants (A) and used most available nutrients (C). However, in the low-appeal condition, the use of blue nutrients—the least costly nutrient—increased across generations. In contrast, blue nutrient use remained stable across generations in the high-appeal condition.

	Plants treated		Num. Blue Nutrients		Num. Nutrients	
(Intercept)	8.41 *** (0.13)	8.53 *** (0.16)	3.81 *** (0.54)	5.00 *** (0.62)	14.96 *** (0.26)	15.02 *** (0.33)
Generation	0.10* (0.04)	0.05 (0.06)	0.36** (0.12)	-0.11 (0.17)	0.14 (0.08)	0.12 (0.11)
Low Appeal	0.00 (0.09)	-0.23 (0.23)	1.84** (0.62)	-0.54 (0.87)	-0.40 (0.23)	-0.52 (0.46)
Generation x Low Appeal		0.09 (0.08)		0.95 *** (0.25)		0.05 (0.16)
Num. obs.	480	480	480	480	480	480
Num. groups: chain	40	40	40	40	40	40

Table A2. Multilevel model estimates for the number of plants treated, the number of blue nutrients used, and the total number of nutrients used in transmitted solutions across treatments and generations. Separate models are reported for each dependent variable using pooled data. All models include random intercepts for transmission chain to account for the hierarchical structure of the data. Standard errors are shown in parentheses and statistical significance is denoted by asterisks (*: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$).

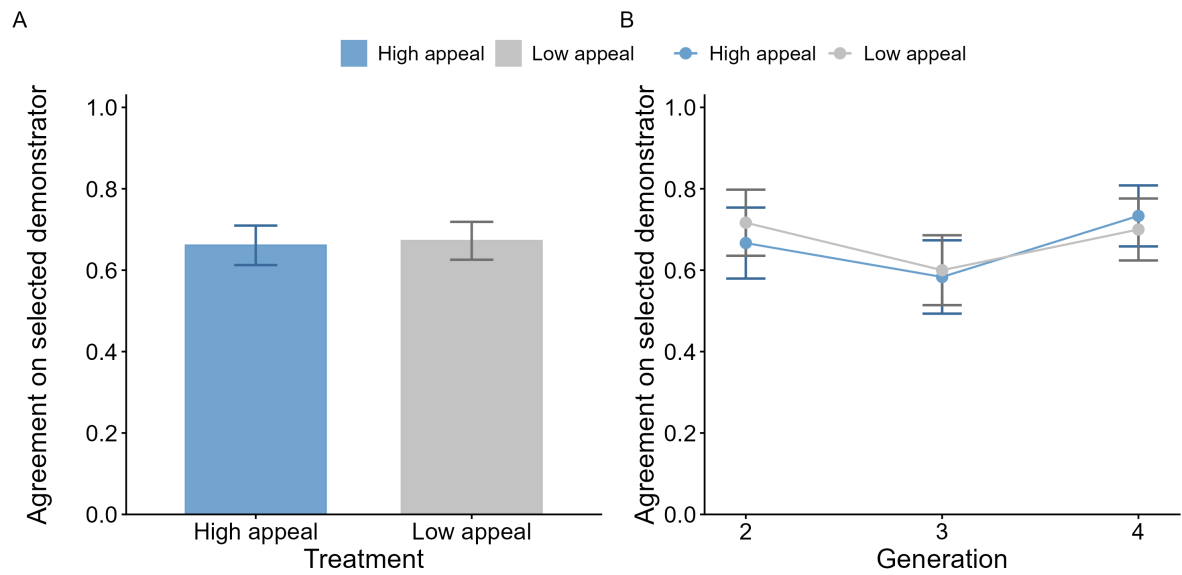


Figure A2. Agreement among learners in demonstrator selection. (A) Mean agreement within each treatment. (B) Mean agreement by generation and treatment. Agreement is computed as the proportion of learner pairs within a chain who chose the same demonstrator. Error bars represent ± 1 SE. Levels of agreement are similar across treatments and remain relatively stable across generations, indicating comparable patterns of demonstrator selection in the high- and low-appeal conditions.

Social Influence

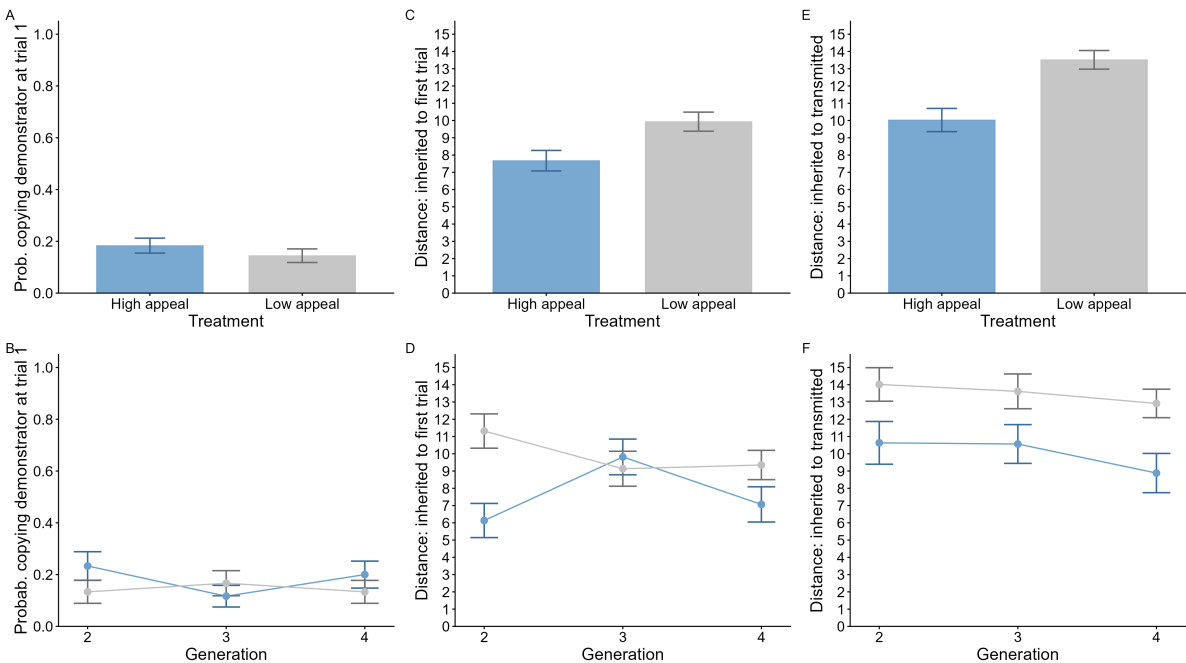
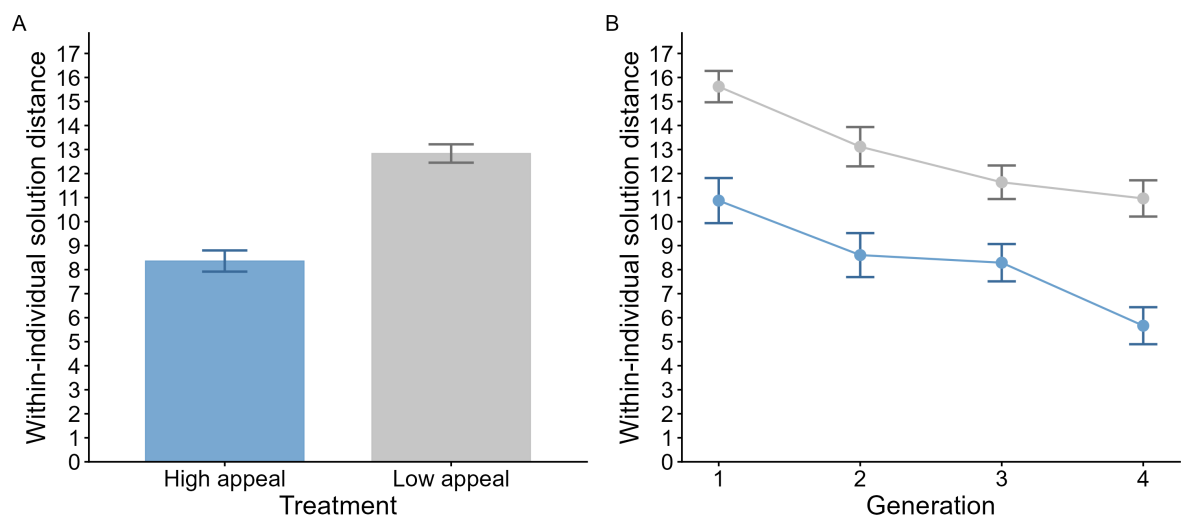


Figure A3. Social influence. (A-B) Probability of copying the inherited solution at Trial 1, defined as the proportion of participants whose first-trial solution exactly matched the inherited solution from their selected demonstrator. (C-D) Mean distance between the inherited solution and the solution used at Trial 1. (E-F) Mean distance between the inherited solution and the transmitted solution (Trial 6). Distance is measured as the sum of absolute differences across all nutrient allocations in the *solution vector*. Panels A, C, and E show treatment-level means, whereas panels B, D, and F show means by treatment and generation. Error bars represent ± 1 SE.

	Probability of copying inherited solution at trial 1			Distance between inherited solution and solution at trial 1			Distance between inherited solution and transmitted solution		
(Intercept)	0.18*** (0.04)	0.21** (0.08)	0.23* (0.10)	7.67*** (0.80)	8.45*** (1.61)	6.27** (2.12)	10.03*** (0.92)	12.17*** (1.71)	12.65*** (2.24)
Low Appeal	-0.04 (0.05)	-0.04 (0.05)	-0.09 (0.15)	2.26* (1.13)	2.26* (1.13)	6.61* (3.00)	3.49* * (1.30)	3.49** (1.30)	2.51 (3.16)
Generation		-0.01 (0.02)	-0.02 (0.03)		-0.26 (0.47)	0.47 (0.66)		-0.71 (0.48)	-0.88 (0.68)
Generatio n x Low Appeal			0.02 (0.05)			-1.45 (0.93)			0.33 (0.96)
Num. obs.	360	360	360	360	360	360	360	360	360
Num. groups: chain	40	40	40	40	40	40	40	40	40

Table A3. Multilevel model estimates for social influence measures: (i) probability of copying the inherited solution at Trial 1, (ii) distance between the inherited solution and the solution used at Trial 1, and (iii) distance between the inherited and transmitted solutions. Separate models are reported for each dependent variable using pooled data. All models include random intercepts for transmission chain to account for the hierarchical structure of the data. Standard errors are shown in parentheses. Statistical significance is denoted by asterisks (*: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$).

613 Individual Exploration



614 **Figure A4 Individual exploration.** (A) Mean within-individual solution distance by treatment. (B) Mean within-
615 individual solution distance by treatment and generation. Distance is measured as the sum of absolute differences
616 across all nutrient allocations in the solution vector averaged over trial pairs. Error bars represent ± 1 SE. Larger
617 distances indicate greater within-individual exploration across trials.
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	Within-individual solution distance	
(Intercept)	12.28*** (0.83)	12.34*** (1.02)
Generation	-1.57*** (0.24)	-1.59*** (0.34)
Low Appeal	4.48*** (0.81)	4.35** (1.44)
Generation x Low Appeal		0.05 (0.48)
Num. obs.	480	480
Num. groups: chain	40	40

622 **Table A4. Multilevel model estimates for individual exploration.** Separate models are reported for each
623 dependent variable (Mean within-individual solution distance) using pooled data. All models include random
624 intercepts for transmission chain to account for the hierarchical structure of the data. Standard errors are shown in
625 parentheses, and statistical significance is denoted by asterisks (*: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$).
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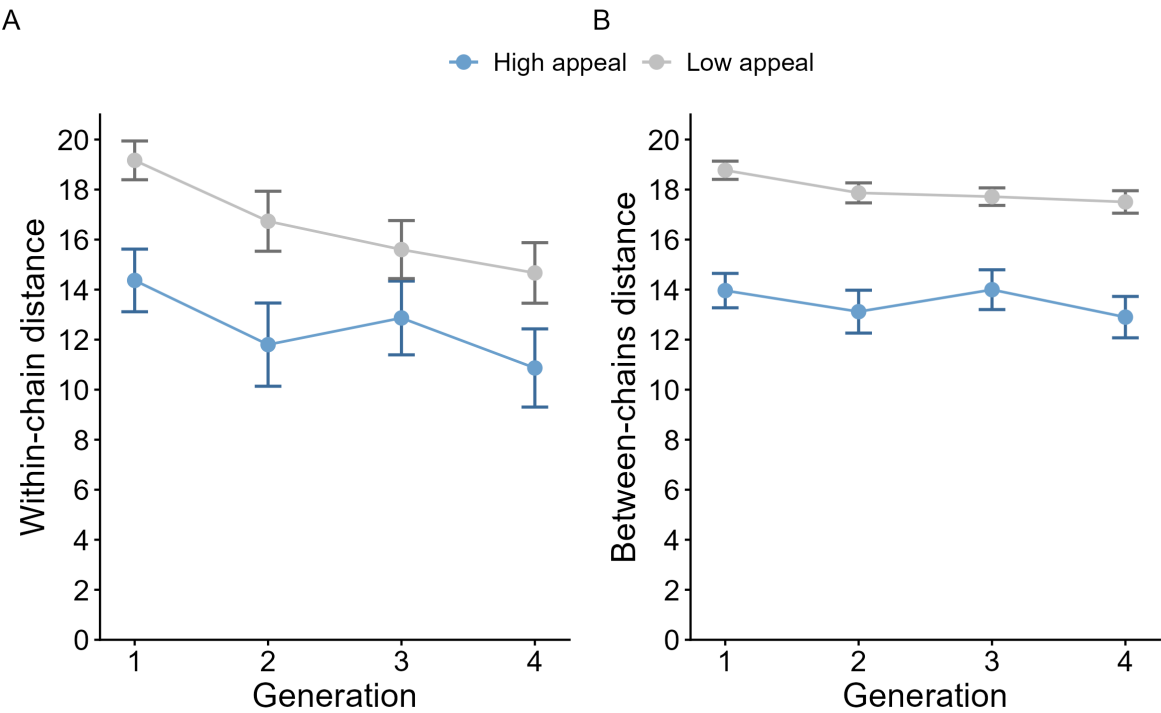


Figure A5. Cultural convergence of transmitted solutions. (A) Within-chain mean pairwise distance among the three transmitted solutions in each chain. (B) Between-chains mean pairwise distance, computed for each chain as the average distance between its three transmitted solutions and all transmitted solutions from other chains in the same treatment-generation group.

	Within-chain solution distance		Between-chain solution distance	
(Intercept)	15.48*** (1.23)	14.83*** (1.57)	14.28*** (0.59)	14.07*** (0.75)
Generation	-1.20** (0.39)	-0.94 (0.56)	-0.31 (0.18)	-0.23 (0.26)
Low Appeal	4.07*** (1.04)	5.37* (2.23)	4.47*** (0.53)	4.88*** (1.06)
Generation x Low Appeal		-0.52 (0.79)		-0.16 (0.37)
Num. obs.	160	160	160	160
Num. groups: chain	40	40	40	40

Table A5. Multilevel model estimates for cultural convergence measures: Within-chain solution distance and between-chain solution distance. Separate models are reported for each dependent variable (within-chain and between-chains solution distance) using pooled data. All models include random intercepts for transmission chain to account for the hierarchical structure of the data. Standard errors are shown in parentheses, and statistical significance is denoted by asterisks (*: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$).

Analyses on Written Advice

Quantitative Textual Analysis

To complement the manual coding of advice content, we conducted quantitative analyses of the linguistic properties of participants' written advice ($n = 480$; 240 per treatment). Before analysis, we removed the fixed prompt prefix ("The field yields a higher output when...") in order to reflect participants' free-response content only. We compared two text measures across treatments and generations: message length, and semantic similarity.

Message Length

Message length was defined as the number of characters in each cleaned advice message. This measure captures the overall verbosity and degree of elaboration in communicated strategies (Figure A6).

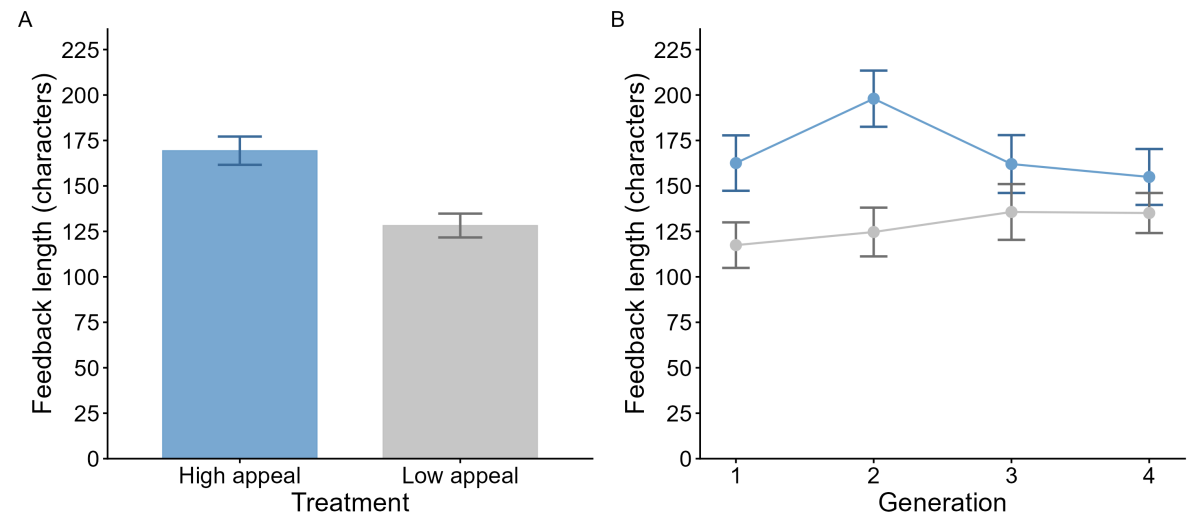


Figure A6. Feedback length in characters generation and treatment. (A) Mean feedback length by treatment. (B) Mean feedback length by treatment and generation. Error bars represent ± 1 SE.

Semantic Similarity

To quantify convergence in the semantic content of advice messages, we used a sentence-embedding approach. Each advice message was transformed into a fixed-length numerical vector using a pre-trained Sentence-BERT (SBERT) model (all-MiniLM-L6-v2). We then computed pairwise cosine similarity between message vectors. Mean cosine similarity was calculated (i) within each treatment overall and (ii) within each treatment $\times$ generation cell by averaging across all pairwise comparisons among messages. This procedure allowed us to examine whether advice became more semantically aligned over generations and whether semantic convergence differed between treatments (Figure A7).

Semantic similarity is initially lower in the low-appeal treatment than in the high-appeal treatment, indicating greater diversity in advice content in early generations. Across generations, similarity increases in the low-appeal condition, whereas changes in the high-appeal condition are comparatively modest. By the fourth generation, both treatments reach a comparable level of within-group semantic similarity.

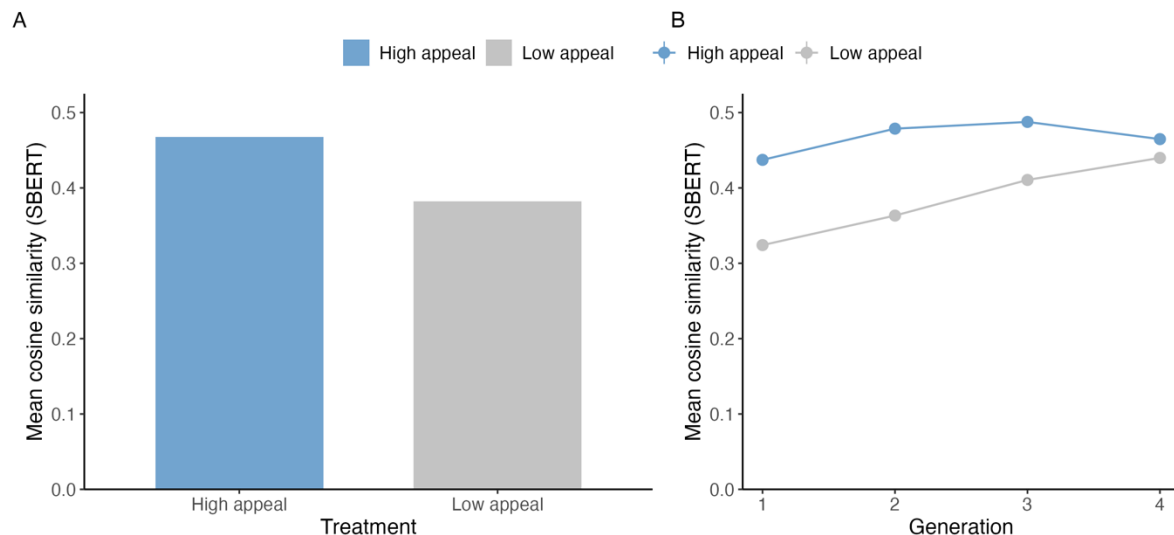


Figure A7. Semantic similarity by treatment and generation. Semantic similarity of transmitted feedback messages using SBERT embeddings and cosine similarity. (A) Overall mean cosine similarity within treatment. (B) Mean cosine similarity within each treatment x generation; each point is the mean over all message pairs within that treatment (and generation). SBERT similarities are from all-MiniLM-L6-v2; higher values indicate greater semantic overlap. Values closer to 1 indicate that two messages occupy nearby positions in semantic space (i.e., express similar underlying ideas), whereas values closer to 0 indicate weaker semantic overlap.

Classification of the written advice

In addition to quantitative outcomes, we analyzed the content of participants' written advice to classify how strategies were represented and transmitted across generations. Written advice text pieces were coded into a set of pre-determined strategy categories. Categories that could conceptually be advised together were treated as non-mutually exclusive, allowing multiple strategy elements to be coded within the same message when applicable. We show category frequencies by generation and treatment condition to assess how transmitted messages evolved over time in Figure A8. We additionally compared the manual classification to an automated LLM-based classification using GPT-5.1. The two approaches agree with 93% accuracy, and all results and figures are robust to using the LLM-based classification (with an exception due to GPT overcategorizing mix nutrients, resulting in 65% accuracy in that category). The classifications, along with the prompt and script used to generate the GPT-5.1 labels, are provided in the supplementary repository.

Category	Description	Example Advice (The field yields a higher output when...)
<i>Mix & Match</i>	Recommends matching nutrient colors to primary plant colors (blue, yellow, red) when possible and combining nutrients to recreate remaining secondary colors (orange, purple, green) based on color theory.	<i>...you match the colours according to colour theory. For example, for green, add one blue and one yellow. For purple, add one red and one blue. and so forth. For the already existing primary colours e.g. red, blue or yellow, just add one of the same colour</i>
<i>Mismatch</i>	Recommends avoiding matching nutrient colors to plant colors (blue, yellow, red), including both simple strategies (using non-corresponding colors) and more complex strategies, such as the opposite of <i>Mix & Match</i> , that allocate remaining	<i>... you use the colours that do not make up the plants colour. If the plant is blue select one each for yellow and red. If the plant is orange, select one for blue as yellow + red = orange so should not be selected.</i>

	primary colors using color theory to contrast the plant's existing color.	
<i>(Other) Plant Color Based matching</i>	Recommends a strategy in which plant color is the primary cue for nutrient allocation (without explicit <i>Mix & Match</i> or <i>Mismatch</i> strategies). Includes simple color matching for primary colors only.	<i>...you focus on matching the colours, I found you got a higher score when prioritising the colours that do not require mixing to match (e.g. giving a red plant two drops of red instead of focusing on giving orange both red and yellow)</i>
<i>Plant Shape Based Matching</i>	Recommends a strategy in which plant shape is the primary cue for nutrient allocation, including feeding plants with the same shape in the same way.	<i>...Crosses are Red, Stars are Yellow I can provide that lower output will be achieved when circles are blue and have tested triangles are blue to some success, Hearts are red.</i>
<i>Even Out/Spread nutrients</i>	Recommends distributing nutrients broadly or evenly across plants or feeding each plant.	<i>...the nutrients are spread out.</i>
Less Nutrients per Plant	Recommends allocating fewer nutrients to each plant without explicitly suggesting even distribution.	<i>...you use fewer bottles of the solution.</i>
More Nutrients per Plant	Recommends allocating more nutrients to each plant.	<i>...you give more than one of the same nutrient per plant.</i>
Mix Nutrient Colors	Recommends combining multiple nutrient colors within a single plant.	<i>...you combine two different solutions on the same field</i>
Pure Nutrients	Recommends avoiding mixing nutrient colors within a plant.	<i>...I used individual colours by themselves rather than combining other colours together</i>
Spatial Strategy	Recommends space-based strategies related to the field (e.g., feeding specific rows, columns, corners, or the center in specific ways).	<i>...blue and yellow is used on the top row. Red and yellow on the middle row.</i>
Use All/Most Nutrients	Recommends using all or nearly all 16 available nutrients per round.	<i>...I used all the nutrients i got a higher score</i>
Time-Based	Recommends strategies based on speed, timing, or temporal pacing.	<i>...you move fast and submit faster</i>
Favor Blue Nutrient	Recommends explicitly prioritizing blue nutrients over other colors (or prioritizing blue equally with another nutrient color).	<i>...you use blue potions</i>
Favor Yellow Nutrient	Recommends explicitly prioritizing yellow nutrients over other colors (or prioritizing yellow equally with another nutrient color).	<i>...when I only use the yellow solution instead of only the blue or red</i>
Favor Red Nutrient	Recommends explicitly prioritizing red nutrients over other colors (or prioritizing red equally with another nutrient color).	<i>...red nutrients are used compared to yellow and blue nutrients</i>
Variation	Recommends diversifying allocations, randomizing, or changing strategies across trials.	<i>...you try to input the colors of the nutrients randomly on different shapes</i>
No Information / No Advice	No Information / No Advice Contains no actionable strategic information or refers only to the latest solution.	<i>...sorry, I really couldn't figure out the pattern for success</i>
Other	Contains advice that does not fit into the predefined categories.	<i>...you trial different combinations, find one that generally scores highly, then repeat that.</i>

Table A6. Coding scheme and illustrative examples of written advice. Advice messages were categorized into strategy types based on their substantive content. Categories are treated as not mutually exclusive; a single message could be assigned to multiple categories where applicable. Descriptions summarize the defining features of each category, and example excerpts illustrate typical coded responses. Examples are extracted from actual participant advice keeping the original spelling and grammatical errors.

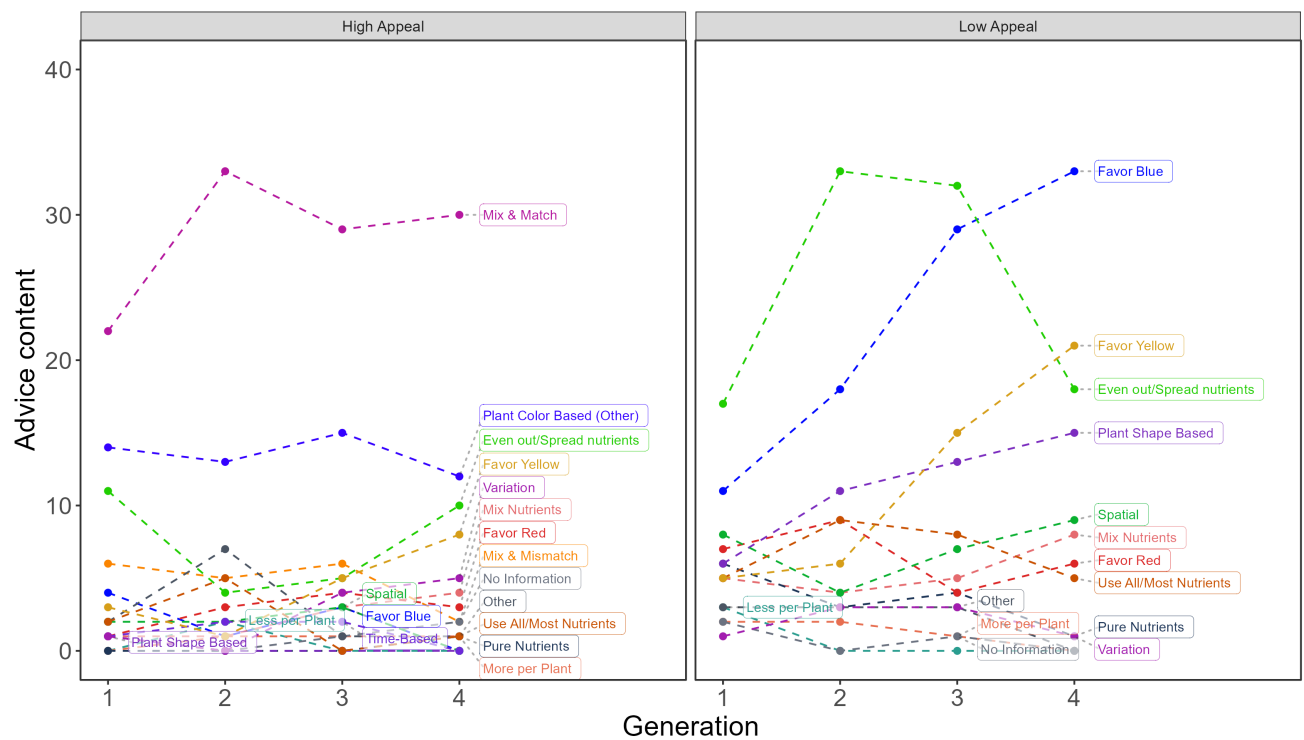


Figure A8. Advice content across generations by treatment conditions. Counts of advice messages coded into a predefined set of strategy categories. In the high-appeal treatment, advice promoting the costly “mix-and-match” strategy increased rapidly in the first two generations and remained relatively stable with a mild decrease. In contrast, in the low-appeal treatment, advice favoring the least costly blue nutrient increased gradually across generations. Advice recommending an even distribution of nutrients—an effective strategy in this task—was more prevalent in the low-appeal treatment.

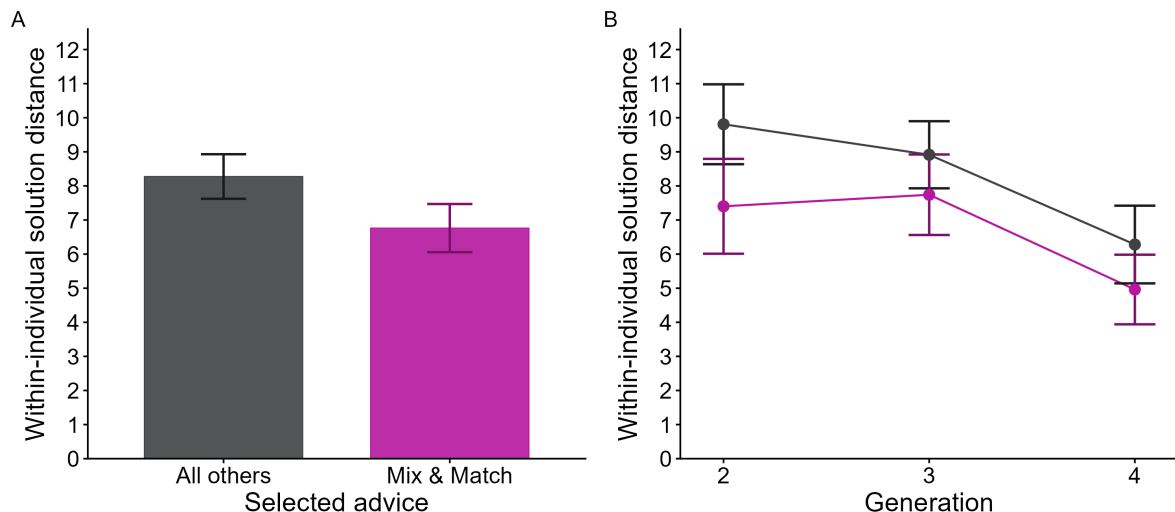


Figure A9. Individual exploration among participants who inherited a mix-and-match solution. (A) Mean within-individual solution distance by treatment. (B) Mean within-individual solution distance by treatment and generation. Distance is measured as the sum of absolute differences across all nutrient allocations in the solution vector, averaged over trial pairs. Error bars represent ± 1 SE. Larger distances indicate greater within-individual exploration across trials.

	Within-individual solution distance	
(Intercept)	12.77*** (1.86)	13.73*** (2.51)
Generation	-1.50** (0.57)	-1.81* (0.80)
Selected Mix & Match	-1.54 (0.95)	-3.50 (3.59)
Generation x Selected Mix & Match		0.66 (1.16)
Num. obs.	180	180
Num. groups: chain	20	20

Table A7. Multilevel model estimates for individual exploration, controlling those who chose to inherit *Mix & Match* solution. All models include random intercepts for transmission chain to account for the hierarchical structure of the data. Standard errors are shown in parentheses and statistical significance is denoted by asterisks (*: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$).

Chain-Level Dynamics

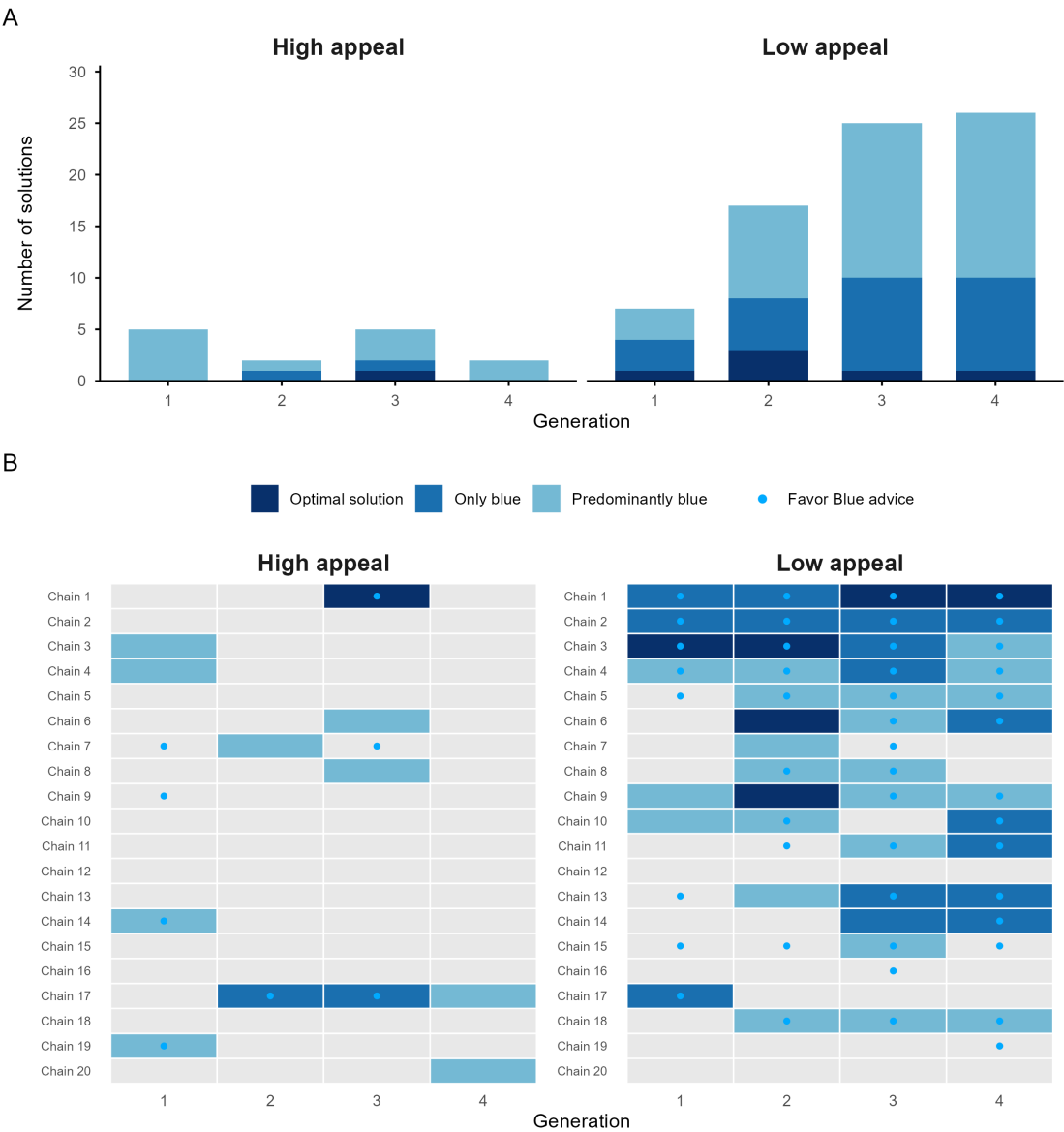


Figure A10. Prevalence of blue-nutrient solutions and written advice favoring blue nutrients. (A) Number of transmitted blue-nutrient heavy solutions per generation, broken down by category: *Optimal solution* (only one blue nutrient for each plant), *Only Blue*, sub-optimal solutions with blue nutrients only (using more than one nutrient each plant or not treating all plants), and solutions that are *blue predominant* (number of blue nutrients is higher than the sum of red and yellow nutrients), for high- and low-appeal conditions. (B) Blue-nutrient status of each chain's transmitted solution at each generation. Each tile reflects the best blue-nutrient strategy solution (Optimal solution > Only blue > Predominantly blue) achieved by any participant within that chain-generation. Dots indicate generations in which at least one participant gave textual advice favoring blue nutrients based on manual classification. Chains are ordered by average expected payoff of the transmitted solution in the last generation within the chain.

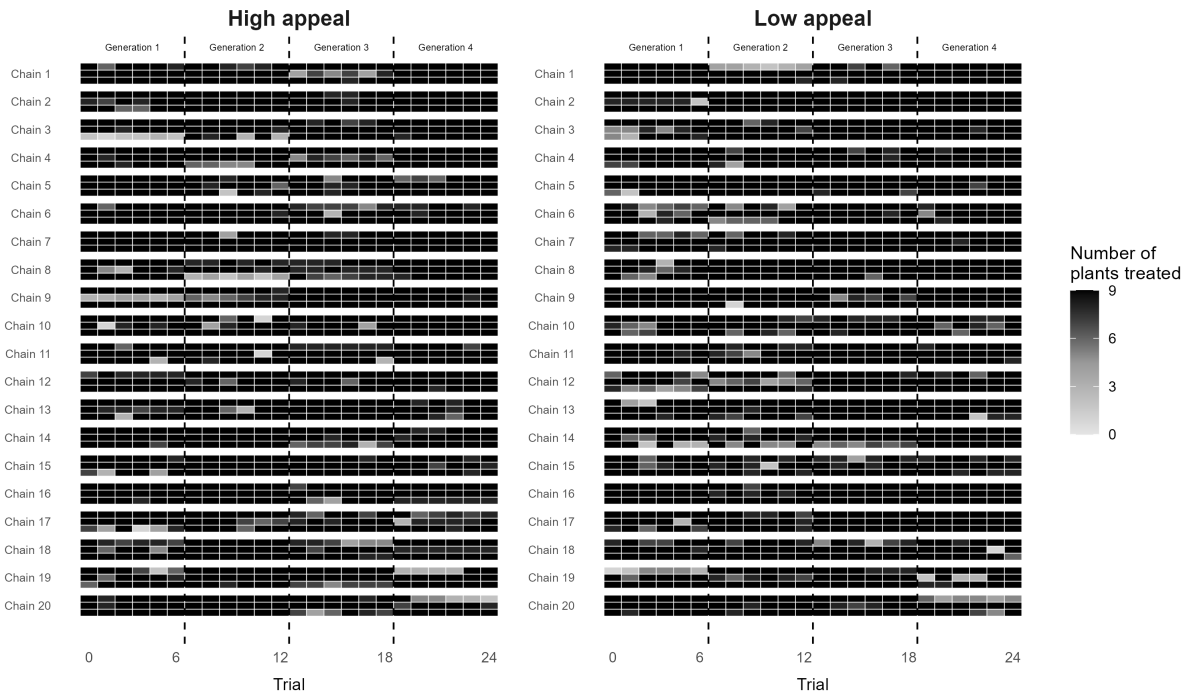


Figure A11. Number of plants treated per trial. Each row represents 4 different generations of participants, and a tile represents a trial by a single participant. Rows are grouped by transmission chains (3 adjacent rows). Color indicates the number of plants treated.

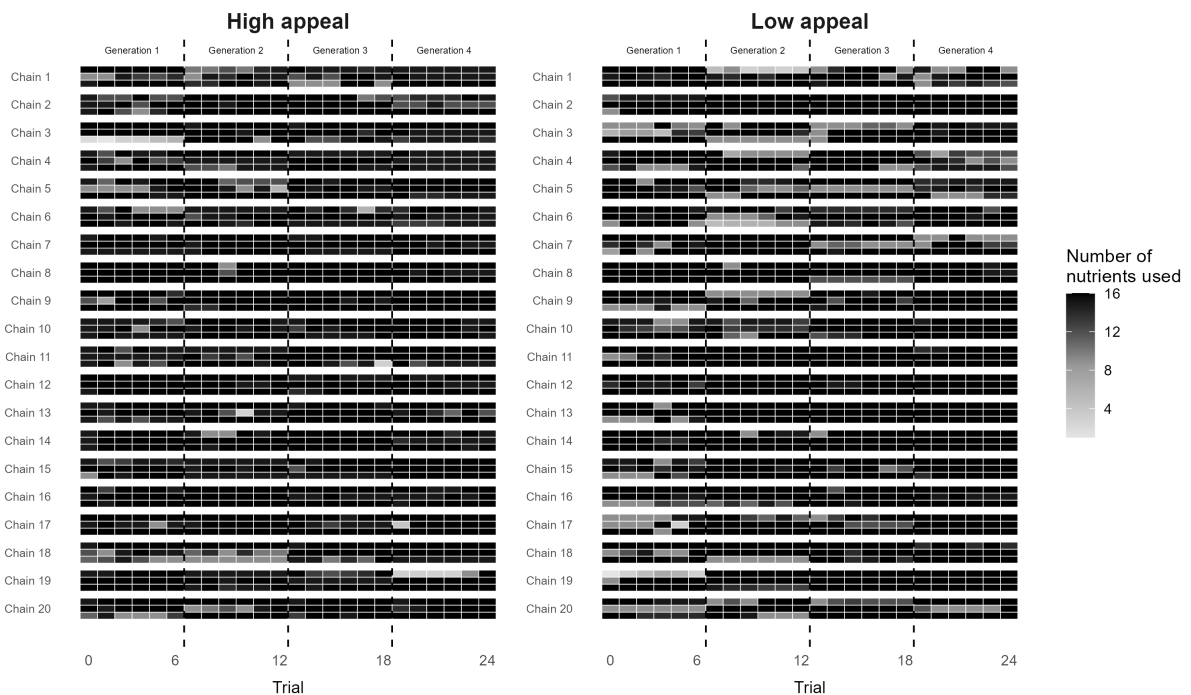


Figure A12. Number of nutrients used per trial. Each row represents 4 different generations of participants, and a tile represents a trial by a single participant. Rows are grouped by transmission chains (3 adjacent rows). Color indicates the number of nutrients used.

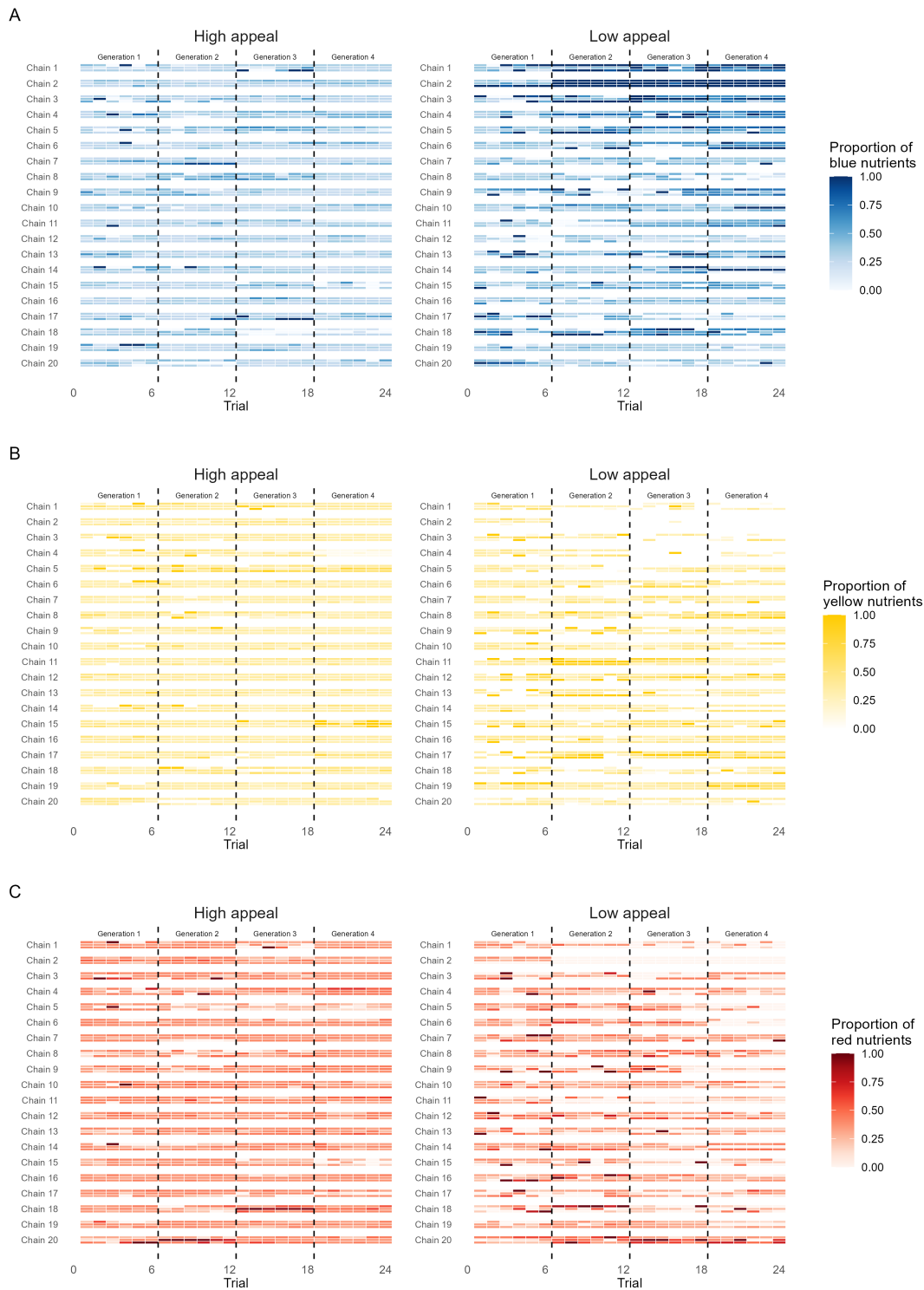


Figure A13. Proportion of each nutrient type per trial. Each row represents 4 different generations of participants, and a tile represents a trial by a single participant. Rows are grouped by transmission chains (3 adjacent rows). (A) Proportion of blue nutrients. (B) Proportion of yellow nutrients. (C) Proportion of red nutrients.

Scores and Costs within Transmission Chains

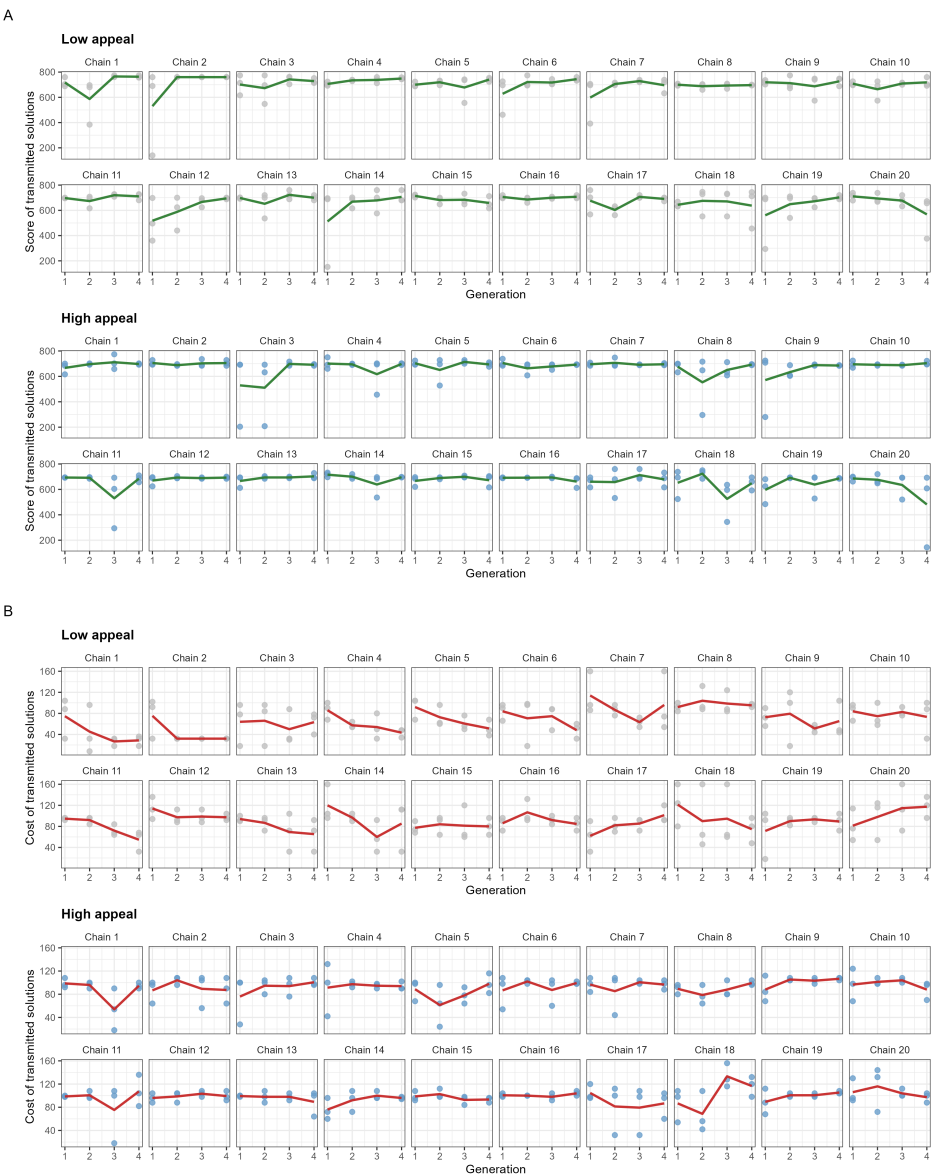


Figure A14. Scores and cost trajectories across chains. (A) Scores (expected) of transmitted solutions within a chain. (B) Costs of transmitted solutions within a chain. Each dot represents a participant's score (or cost) of the transmitted solution. Green lines indicate the average score, and red lines indicate the average cost within the chain.

Cumulative Cultural Evolution without Payoff-Biased Social Learning

In a previous study, we investigated a variant of this experiment that was identical to the present design except that participants were not provided with information about demonstrators' scores. Participants instead relied exclusively on written advice transmitted by earlier participants prior to selecting a solution. This design allowed for social learning via instruction transmission, while eliminating the possibility of payoff-biased social learning.

Results show no statistically significant evidence of cumulative improvement in scores across generations in either treatment. Although the figures suggest a gradual divergence in solution costs between treatments (Figure A15), this difference is not statistically supported by the regression analyses (Table A8). We see a similar pattern of exploration in this variant of the game: In Low-Appeal treatment, participants create a higher number of unique solutions (Figure A16).

Together, these findings suggest that in this task, payoff-biased social learning is necessary but not sufficient for cumulative cultural evolution. While payoff information can overcome payoff noise and enable adaptive refinement in environments with low cognitive appeal, strong biases in expectations of efficacy can erode its benefits, limit the exploration and allow inefficient solutions to persist even under conditions typically favorable to cumulative improvement.

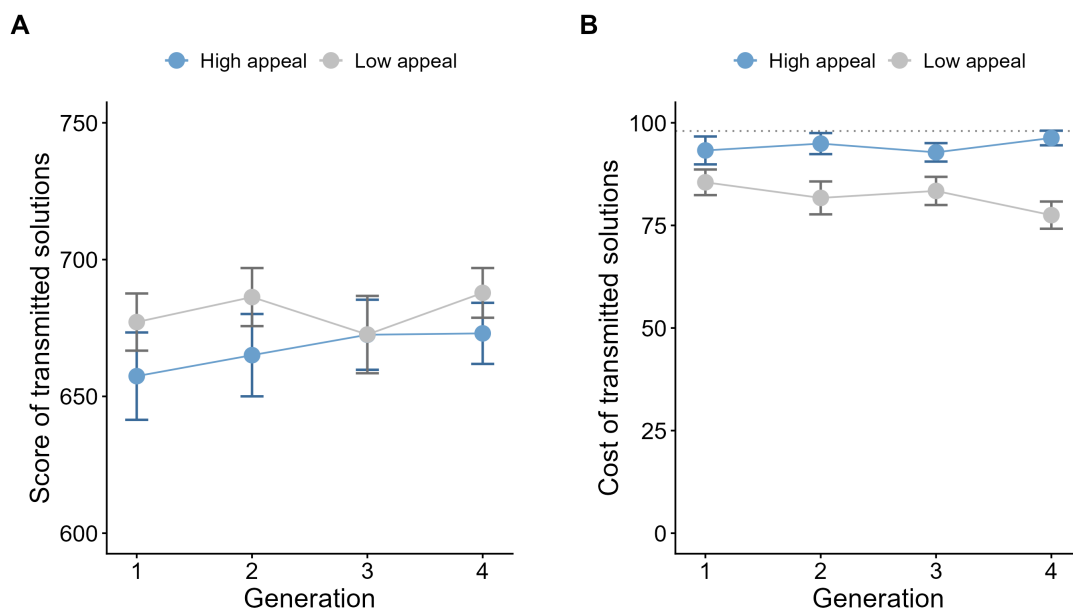


Figure A15. Study without demonstrator payoff information: Score and cost of culturally transmitted solutions across generations. A) In the low-appeal treatment (gray line), participants showed gradual improvements in solution scores across generations, whereas scores in the high-appeal treatment (blue line) did not change significantly. B) Participants in the low-appeal treatment progressively reduced the cost of their solutions, while those in the high-appeal treatment showed no statistically significant change over time. The dotted line represents the cost of the mix-and-match solution (see Figure 5 for an illustration).

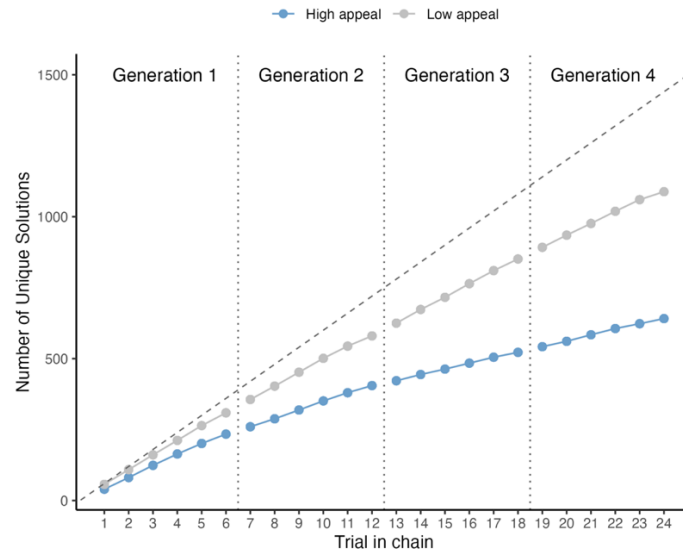


Figure A16. Study without demonstrator payoff information: Exploration across trials. Cumulative number of unique solutions (i.e., nutrient-to-plant allocation patterns) across trials in each treatment. The dashed black line indicates the theoretical upper bound of exploration, assuming every participant submitted a solution not previously produced in this treatment. The gray line shows the cumulative number of unique solutions observed in the low-appeal treatment, while the blue line shows the high-appeal treatment.

A) Scores			
	Low-Appeal only	High-Appeal only	Pooled
(Intercept)	676.40 *** (13.92)	653.42 *** (16.92)	653.42 *** (15.48)
Generation	1.83 (4.95)	5.44 (6.18)	5.44 (5.60)
Low Appeal			22.98 (21.90)
Generation x Low Appeal			-3.61 (7.92)
B) Costs			
(Intercept)	87.60 *** (4.60)	92.58 *** (3.22)	92.58 *** (3.97)
Generation	-2.23 (1.43)	0.70 (1.11)	0.70 (1.28)
Low Appeal			-4.98 (5.62)
Generation x Low Appeal			-2.93 (1.81)
Num. obs.	240	240	480
Num. groups: chain	20	20	40

Table A8. Study without demonstrator payoff information: Multilevel model estimates for expected scores (A) and costs (B) of culturally transmitted solutions across treatments. Separate models are reported for the low-appeal condition, high-appeal condition, and pooled data to compare changes in expected scores/costs over generations, with random intercepts controlling for transmission *chain* structure of the experiment. Standard errors are shown in parentheses and statistical significance is denoted by asterisks (*: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$).

Experimental Instructions and Task Screenshots

Instructions and control questions

1/12

Welcome to our study

Over the next few slides, we will explain the rules of the game and what you must do to maximize your monetary reward. Read them carefully so that you can earn as much bonus payment as possible.

Next

High Appeal

2/12

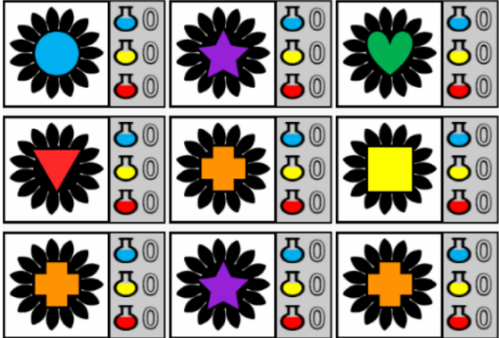
Principle of the game

Your goal is to maximize the output of a field of 9 plants across 6 rounds.

Plants come in a variety of forms and colors and can be fed using 3 distinct nutrients.

The way you feed your plants affects the output of your field. However, we will not tell you how best to feed your plants. This is for you to figure out.

Previous



Next

Low Appeal

2/12

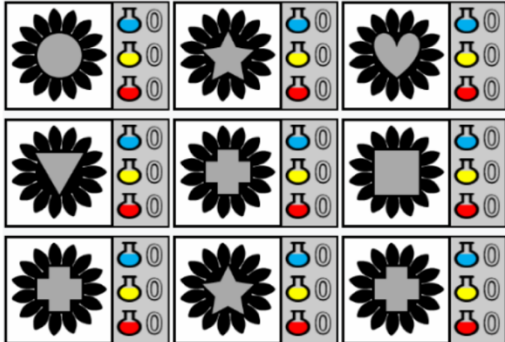
Principle of the game

Your goal is to maximize the output of a field of 9 plants across 6 rounds.

Plants come in a variety of forms and can be fed using 3 distinct nutrients.

The way you feed your plants affects the output of your field. However, we will not tell you how best to feed your plants. This is for you to figure out.

Previous



Next

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High Appeal

3/12

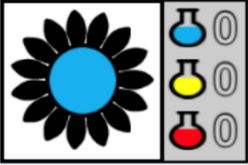
Adding nutrients to plants

Different nutrients have different colors. In each round, you can add several droplets of nutrients to each plant. You can also mix nutrients.

Each plant has a control panel that allows you to provide droplets of nutrients to that plant. Hover your mouse over the nutrients to reveal the plant control panel.

Try it out! Feed your plant with 1 red nutrient and leave the other nutrients at 0.

Previous



You need to complete the task before continuing.

822

823

824

Low Appeal

3/12

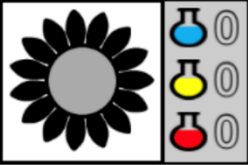
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Try it out! Feed your plant with 1 red nutrient and leave the other nutrients at 0.

Previous



You need to complete the task before continuing.

825

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High Appeal

4/12

Earning scores

At each round, you must use a total number of droplets between 1 and 16 to feed all the plants in your field. **The way you feed your plants will determine your payoff.** However, it will also depend on an element of chance. In other words, even if you choose the same solution twice, there is no guarantee that the output will be exactly the same.

Previous round

Score in the previous round

points

Your solution in the previous round:





































Preview of the results after each round.
(Actual values are obfuscated.)

Previous

Next

829
830
831

Low Appeal

4/12

Earning scores

At each round, you must use a total number of droplets between 1 and 16 to feed all the plants in your field. **The way you feed your plants will determine your payoff.** However, it will also depend on an element of chance. In other words, even if you choose the same solution twice, there is no guarantee that the output will be exactly the same.

Previous round

Score in the previous round

points

Your solution in the previous round:





































Preview of the results after each round.
(Actual values are obfuscated.)

Previous

Next

832
833
834

5/12

Please answer the questions based on the provided information. You are allowed to go back and re-read the information.

What factors determine your score in each round?

The way the field has been fed

An element of chance

Both of the above

Previous

You need to complete the task before continuing.

6/12

Please answer the questions based on the provided information. You are allowed to go back and re-read the information.

How many nutrients can you use in total in each round?

Any number

Any number from 1 to 16

Exactly 16

Previous

You need to complete the task before continuing.

7/12

Collective task and social information

Improving the output of the field is a collective task. Although you are playing alone, you are part of a chain of several players, each of whom also has 6 rounds to improve their solution.

You are the first player in the chain. In the last round, we will ask you to write a piece of advice for the next player and share your solution.

The number of players in a chain is not disclosed. So, there might be, or might not be, a player after you.



Previous

Next

Interface

On the right side of your screen, you will be able to input your solution using the control panels. At any time, you can see the number of nutrients you would like to use in the current round.

Preview of the interface.

Next

Interface

On the right side of your screen, you will be able to input your solution using the control panels. At any time, you can see the number of nutrients you would like to use in the current round.

Preview of the interface.

Next

844
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846
847

Low Appeal, 2nd-4th generation

8/12

Interface

On the left side of your screen, a tab will provide you with information about the previous round of the game. In a second tab, you will see the advice and solution from a previous player.

On the right side of your screen, you will be able to input your solution using the control panels. At any time, you can see the number of nutrients you would like to use in the current round.

Previous round (You)

Advice from the previous player

Score in the previous round

points

Your solution in the previous round:

Current round:

2 / 6

Total score:

points

Select your solution for this round:

You used 0 nutrients.

Submit your solution

Preview of the interface.

Previous

Next

848
849
850

9/12

Monetary reward

Your reward will depend on how well you did in the game and how well a future player does in their first round. We will add up the points you obtained in each round. Then, we will add the points earned by the next player in their first round to your points (unless you are the last participant in the chain).

We will then convert those points into a monetary reward in GBP. The conversion rate is:

2500 points = £1.

This reward will be added to your participation fee on Prolific as a bonus payment.

Previous

Next

851

10/12

Please answer the questions based on the provided information. You are allowed to go back and re-read the information.

How is your bonus payment calculated?

It will be based only on my own score.

It will be based on the score of the next player in the chain.

It will be based on my own score and the score of the next player in the first round.

Previous

You need to complete the task before continuing.

11/12

Please answer the questions based on the provided information. You are allowed to go back and re-read the information.

What is the equivalent of 2500 points in pounds?

£1

£1.5

£0.5

Previous

You need to complete the task before continuing.

12/12

You have finished the instructions. You won't be able to go back to them once you start the experiment, so please make sure you have understood everything before starting and go through the instructions again if you feel you need to.

To start the experiment, click on the "Start Experiment" button below.

Previous

Start Experiment

858 Screenshots from the gameplay

859
860

Task Screen, Low Appeal, First generation

Previous round

Score in the previous round321 points

Your solution in the previous round:

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Current round:2 / 6

Total score:321 points

Select your solution for this round:

You used 11 nutrients.

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Submit your solution

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Task screen, Last round, High Appeal, First generation

Previous round

Score in the previous round665 points

Your solution in the previous round:

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Current round:6 / 6

Total score:1357 points

This is your last trial.

This solution that you will produce at this round will be transmitted to the next participant along with a message you will write in the next step and the score you will earn from this round.

Remember that your final monetary reward will depend on how that player performs in their first round. In order to help future participants and to maximise your own reward, it is important that your demonstration is informative.

Select your solution for this round:

You used 0 nutrients.

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Submit your solution

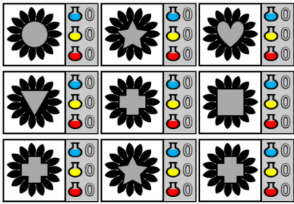
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870

Advice Selection Screen, Low Appeal, 2nd-4th generation

Here is the field that you will have to feed. Below it are three pieces of advice from previous players who have completed 6 rounds.

Select a player to be provided with their advice and solution for the entire duration of the experiment.



Demonstration of the field.

Player 1

Score at the last trial:
706 points

Advice written:

The field yields a higher output when you keep the same entry every round.

Select this advice

Player 2

Score at the last trial:
477 points

Advice written:

The field yields a higher output when you miss at least one completely and then for a few around 1-3 have 1 at least for each colour

Select this advice

Player 3

Score at the last trial:
850 points

Advice written:

The field yields a higher output when you use all up the droplets. Some shapes perform better with different colours. the cross performed better with yellow

Select this advice

871
872
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Task Screen, Low Appeal, 2nd-4th generation

Previous round (You)

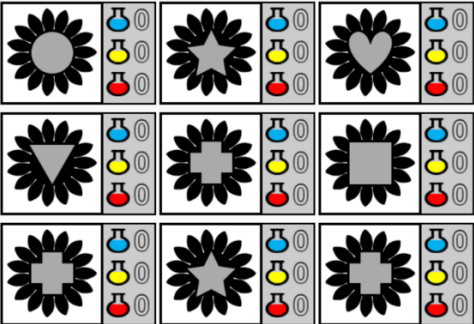
Advice from the previous player

Current round: 1 / 6

Total score: 0 points

Select your solution for this round:

You used 0 nutrients.

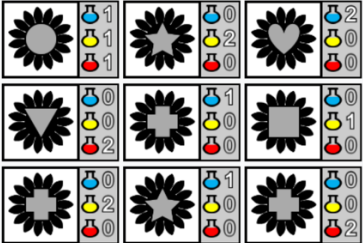


Submit your solution

Advice from the previous player: *The field yields a higher output when you use all up the droplets. Some shapes perform better with different colours. the cross performed better with yellow*

Solution provided by the previous player:

Score: 850 points



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Task Screen, Low Appeal, 2nd-4th generation

Previous round (You)

Advice from the previous player

Current round: 2 / 6

Total score: 587 points

Score in the previous round

587 points

Select your solution for this round:

You used 0 nutrients.

Your solution in the previous round:



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Submit your solution

891
892
893

Advice Message Screen, High Appeal

Previous round

This solution will be transmitted to the next player.
You will see the score of this round at the end of the experiment.

Please complete the following sentence to transmit your advice to the next player. Remember that your final monetary reward will depend on how that player performs in their first round. In order to help future participants and to maximise your own reward, it is important that your advice is clear and informative.

The field yields a higher output when

Submit your advice



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